

P25 J3399: Non-uniform Refinement

CryoSPARC Running Version: v4.7.1

Project Details

DETAIL	VALUE
Title	24sep20b_WT-Control-White_From-January
Description	This is a repeat collection of White Control Wildtype CFTR from the original January/February 2024 collection. Using
ID	P25
Created by user	Andy Paige

Job Details

DETAIL	VALUE
Title	New Job J3399
Description	Enter a description.
ID	J3399
Job type	Non-uniform Refinement
Status	completed
Created by user	Minou Emmad
Created	Tue Jun 09 2026 15:14:55 GMT-0400 (Eastern Daylight Time)
Queued	Tue Jun 09 2026 15:15:14 GMT-0400 (Eastern Daylight Time)
Launched	Tue Jun 09 2026 15:17:20 GMT-0400 (Eastern Daylight Time)
Started	Tue Jun 09 2026 15:17:22 GMT-0400 (Eastern Daylight Time)
Waiting	-
Killed	-
Completed	Tue Jun 09 2026 15:39:12 GMT-0400 (Eastern Daylight Time)
Failed	-
Last accessed	Wed Jun 10 2026 15:56:45 GMT-0400 (Eastern Daylight Time)
Size	6.13 GB

Inputs

particles

J3394.particles_all_classes

BLOB
J3394.particles_all_classes.blob.F
CTF
J3394.particles_all_classes.ctf.F

ALIGNMENTS3D

J3394.particles_all_classes.alignments3D.F
PASSTHROUGH
J3394.particles_all_classes.alignments3D_multi.F

volume

J3394.volume_class_0
MAP
J3394.volume_class_0.map.F

mask

No group connections

Parameters

Particle preprocessing

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED
Window dataset (real-space)	True	X		
Window inner radius	0.85	X		X
Window outer radius	0.99	X		X

Homogeneous Refinement

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED
Symmetry	C1	X		
Symmetry relaxation method	none	X		X
Do symmetry alignment	True	X		X
Re-estimate greyscale level of input reference	True	X		X
Number of extra final passes	0	X		X
Maximum align resolution (A)	None	X		
Initial lowpass resolution (A)	30	X		X
GSFSC split resolution (A)	20	X		X
Force re-do GS split	False	X		X
Adaptive Marginalization	True	X		X
Non-uniform refine enable	True	X		X
Non-uniform filter order	8	X		X
Non-uniform AWF	3	X		X
Enforce non-negativity	False	X		X
Window structure in real space	True	X		X
Skip interpolant premult	True	X		X
Ignore DC component	True	X		X
Ignore tilt	False	X		X

Ignore trefoil	False	X		X	
Ignore tetra	False	X		X	
Use random ordering of particles	True	X		X	
Disable auto batchsize	False	X		X	
Batchsize epsilon	0.001	X		X	
Batchsize snrfactor	50	X		X	
Reset input per-particle scale	True	X		X	
Minimize over per-particle scale	False	X		X	
Scale min/use start iter	0	X		X	
Noise priorw	50	X		X	
Noise initw	200	X		X	
Noise initial sigma-scale	3	X		X	
Initialize noise model from images	False	X		X	
Dynamic mask threshold (0-1)	0.2	X		X	
Dynamic mask near (A)	6	X		X	
Dynamic mask far (A)	14	X		X	
Dynamic mask start resolution (A)	12	X			
Dynamic mask use absolute value	False	X		X	
Show plots from intermediate steps	True	X			
GPU batch size of images	None	X		X	

Defocus Refinement

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Optimize per-particle defocus	False	X		X	
Num. particles to plot	3	X			
Minimum Fit Res (A)	20	X			
Defocus Search Range (A +/-)	2000	X			
GPU batch size of images	None	X		X	

Global CTF Refinement

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Optimize per-group CTF params	False	X		X	
Num. groups to plot	3	X			
Binning to apply to plots	1	X		X	
Minimum Fit Res (A)	10	X			
Fit Tilt	True	X			
Fit Trefoil	True	X			

Fit Spherical Aberration	False	X			
Fit Tetrafoil	False	X			
Fit Anisotropic Mag.	False	X			
GPU batch size of images	None	X		X	

Ewald Sphere Correction

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Do EWS correction	False	X		X	
EWS curvature sign	negative	X		X	

Random Seeds

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Random seed	99029637	X		X	

Compute settings

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Cache particle images on SSD	True	X			
Low-Memory Mode	False	X			

Outputs

particles

ALIGNMENTS3D	
J3399.particle.alignments3D	

CTF	
J3399.particle.ctf	

BLOB (PASSTHROUGH)	
J3399.particle.blob	

ALIGNMENTS3D_MULTI (PASSTHROUGH)	
J3399.particle.alignments3D_multi	

volume

MAP	
J3399.volume.blob	

MAP_SHARP	
J3399.volume.blob	

MAP_HALF_A	

J3399.volume.blob	
MAP_HALF_B	
J3399.volume.blob	
MASK_REFINE	
J3399.volume.blob	
MASK_FSC	
J3399.volume.blob	
MASK_FSC_AUTO	
J3399.volume.blob	
PRECISION	
J3399.volume.blob	

mask

MASK_REFINE	
J3399.volume.blob	

Target

ganges.biology.columbia.edu (node)

```
> [2026-06-09 15:17:20.62 GMT-4] License is valid.
> [2026-06-09 15:17:20.63 GMT-4] Launching job on lane default target ganges.biology.columbia.edu ...
> [2026-06-09 15:17:20.64 GMT-4] Running job on master node hostname ganges.biology.columbia.edu
> [2026-06-09 15:17:22.64 GMT-4] [CPU: 78.9 MB]      [Avail: 316.59 GB] Job J3399 Started
> [2026-06-09 15:17:22.65 GMT-4] [CPU: 78.9 MB]      [Avail: 316.56 GB] Master running v4.7.1, worker running v4.7.1
> [2026-06-09 15:17:22.67 GMT-4] [CPU: 79.2 MB]      [Avail: 316.54 GB] Working in directory:
                                /mnt/nfs_HuntLabSynologyDS1621CryoEM/hCFTR/Movies/24sep20b/24
                                sep20b/rawdata/CS-24sep20b-wt-control-white-from-january/J339
                                9
> [2026-06-09 15:17:22.68 GMT-4] [CPU: 79.2 MB]      [Avail: 316.54 GB] Running on lane default
> [2026-06-09 15:17:22.68 GMT-4] [CPU: 79.2 MB]      [Avail: 316.54 GB] Resources allocated:
> [2026-06-09 15:17:22.68 GMT-4] [CPU: 79.2 MB]      [Avail: 316.53 GB]   Worker:  ganges.biology.columbia.edu
> [2026-06-09 15:17:22.69 GMT-4] [CPU: 79.2 MB]      [Avail: 316.53 GB]   CPU    :  [6, 7, 14, 15]
> [2026-06-09 15:17:22.69 GMT-4] [CPU: 79.2 MB]      [Avail: 316.53 GB]   GPU    :  [2]
> [2026-06-09 15:17:22.69 GMT-4] [CPU: 79.2 MB]      [Avail: 316.53 GB]   RAM    :  [3, 9, 10]
> [2026-06-09 15:17:22.69 GMT-4] [CPU: 79.2 MB]      [Avail: 316.53 GB]   SSD    :  True
> [2026-06-09 15:17:22.70 GMT-4] [CPU: 79.2 MB]      [Avail: 316.53 GB] -----
                                -
> [2026-06-09 15:17:22.70 GMT-4] [CPU: 79.2 MB]      [Avail: 316.54 GB] Importing job module for job type nonuniform_refine_new...
> [2026-06-09 15:17:30.76 GMT-4] [CPU: 273.6 MB]     [Avail: 316.18 GB] Job ready to run
> [2026-06-09 15:17:30.76 GMT-4] [CPU: 273.6 MB]     [Avail: 316.18 GB] *****
                                **
> [2026-06-09 15:17:31.49 GMT-4] [CPU: 296.9 MB]     [Avail: 316.10 GB] Using random seed of 99029637
> [2026-06-09 15:17:31.50 GMT-4] [CPU: 296.9 MB]     [Avail: 316.13 GB] Loading a ParticleStack with 15925 items...
> [2026-06-09 15:17:31.51 GMT-4] [CPU: 296.9 MB]     [Avail: 316.22 GB]

                                SSD cache ACTIVE at
                                /scr/instance_ganges.biology.columbia.edu:39001 (200 GB
                                reserve) (3 TB quota)

                                Cache usage                                Amount

                                Total / Usable                            3.58 TiB / 2.73 TiB
                                Used / Free                              2.73 TiB / 666.66 MiB
                                Hits / Misses                             73.29 GiB / 0.00 B
                                Acquired / Required                      73.29 GiB / 0.00 B

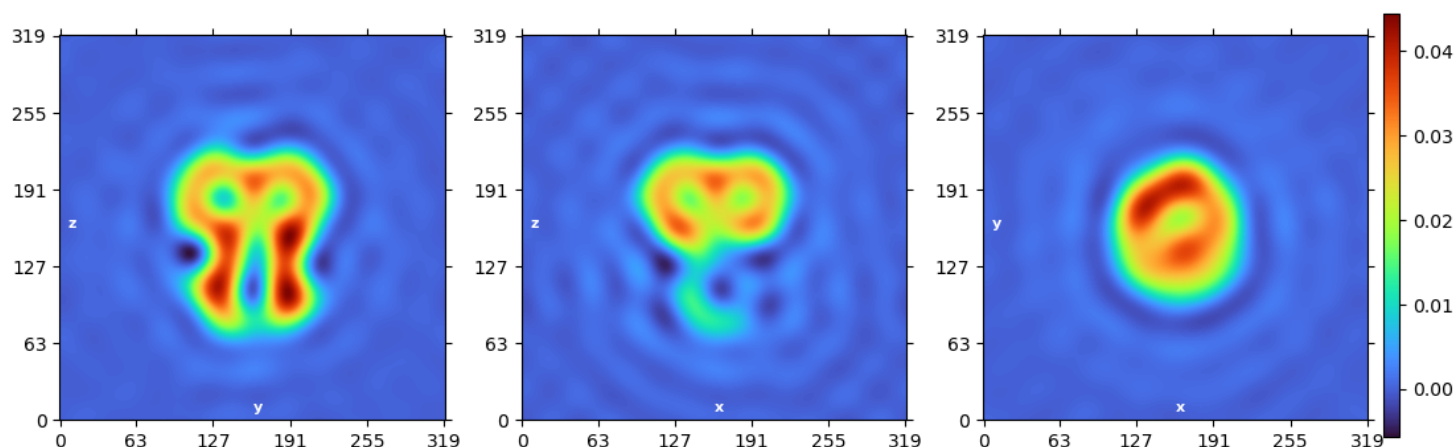
                                Progress:
                                [
                                7390/7390 (100%)
                                Elapsed: 0h 00m 06s
                                Active jobs: P25-J3398, P25-J3397, P25-J3399
                                SSD cache complete for 7390 file(s)

> [2026-06-09 15:17:39.04 GMT-4] [CPU: 348.5 MB]     [Avail: 314.73 GB] Done.
> [2026-06-09 15:17:39.04 GMT-4] [CPU: 348.5 MB]     [Avail: 314.71 GB] Windowing particles
> [2026-06-09 15:17:39.05 GMT-4] [CPU: 348.8 MB]     [Avail: 314.68 GB] Done.
> [2026-06-09 15:17:39.05 GMT-4] [CPU: 348.8 MB]     [Avail: 314.67 GB] ===== Gold Standard Split =====
> [2026-06-09 15:17:39.06 GMT-4] [CPU: 348.8 MB]     [Avail: 314.66 GB] Particles have input alignments3D connected, so reusing
                                pre-existing split
> [2026-06-09 15:17:39.06 GMT-4] [CPU: 348.8 MB]     [Avail: 314.64 GB] Set A is smaller than set B by 19 particles (0.119 percent
                                difference relative to the total dataset).
> [2026-06-09 15:17:39.14 GMT-4] [CPU: 349.9 MB]     [Avail: 314.40 GB] Split A has 7953 particles
> [2026-06-09 15:17:39.15 GMT-4] [CPU: 349.9 MB]     [Avail: 314.39 GB] Split B has 7972 particles
> [2026-06-09 15:17:39.15 GMT-4] [CPU: 349.9 MB]     [Avail: 314.38 GB] ===== Refinement =====
> [2026-06-09 15:17:39.15 GMT-4] [CPU: 349.9 MB]     [Avail: 314.37 GB] Input particles have box size 320
> [2026-06-09 15:17:39.16 GMT-4] [CPU: 349.9 MB]     [Avail: 314.36 GB] Input particles have pixel size 0.8300
> [2026-06-09 15:17:39.16 GMT-4] [CPU: 349.9 MB]     [Avail: 314.35 GB] Particles will be zeropadded/truncated to size 320 during
                                alignment
```

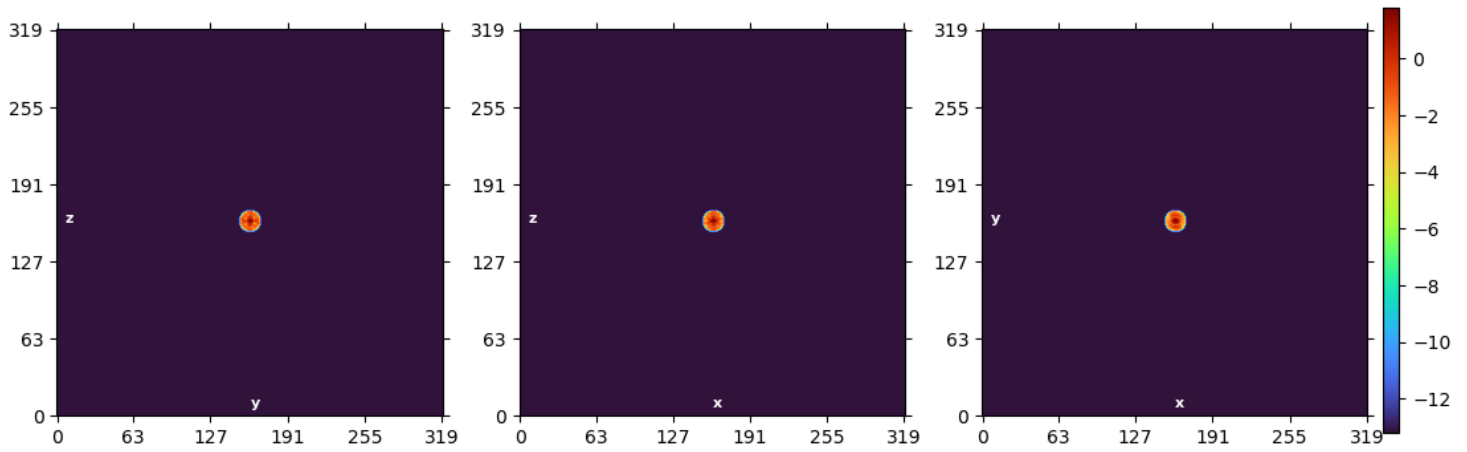
```

> [2026-06-09 15:17:39.16 GMT-4] [CPU: 349.9 MB] [Avail: 314.34 GB] Volume refinement will be done with effective box size 320
> [2026-06-09 15:17:39.17 GMT-4] [CPU: 349.9 MB] [Avail: 314.33 GB] Volume refinement will be done with pixel size 0.8300
> [2026-06-09 15:17:39.17 GMT-4] [CPU: 349.9 MB] [Avail: 314.32 GB] Particles will be zeropadded/truncated to size 320 during
backprojection
> [2026-06-09 15:17:39.17 GMT-4] [CPU: 349.9 MB] [Avail: 314.30 GB] Particles will be backprojected with box size 320
> [2026-06-09 15:17:39.17 GMT-4] [CPU: 350.0 MB] [Avail: 314.29 GB] Volume will be internally cropped and stored with box size
320
> [2026-06-09 15:17:39.18 GMT-4] [CPU: 350.0 MB] [Avail: 314.28 GB] Volume will be interpolated with box size 320 (zeropadding
factor 1.00)
> [2026-06-09 15:17:39.18 GMT-4] [CPU: 350.0 MB] [Avail: 314.27 GB] DC components of images will be ignored and volume will be
floated at each iteration.
> [2026-06-09 15:17:39.18 GMT-4] [CPU: 350.0 MB] [Avail: 314.26 GB] Spherical windowing of maps is enabled
> [2026-06-09 15:17:39.19 GMT-4] [CPU: 350.1 MB] [Avail: 314.24 GB] Refining with C1 symmetry enforced
> [2026-06-09 15:17:39.19 GMT-4] [CPU: 350.1 MB] [Avail: 314.23 GB] Refining with pose and shift marginalization during
backprojection enabled.
> [2026-06-09 15:17:39.19 GMT-4] [CPU: 350.1 MB] [Avail: 314.22 GB] Resetting input per-particle scale factors to 1.0
> [2026-06-09 15:17:39.20 GMT-4] [CPU: 350.1 MB] [Avail: 314.21 GB] Starting at initial resolution 30.000A (radwn 8.853).
> [2026-06-09 15:17:39.20 GMT-4] [CPU: 350.1 MB] [Avail: 314.20 GB] ===== Non-Uniform Refinement =====
> [2026-06-09 15:17:39.20 GMT-4] [CPU: 350.1 MB] [Avail: 314.20 GB] Non-Uniform Refinement is enabled.
> [2026-06-09 15:17:39.21 GMT-4] [CPU: 350.1 MB] [Avail: 314.19 GB] Using AWF of 3.00.
> [2026-06-09 15:17:39.21 GMT-4] [CPU: 350.1 MB] [Avail: 314.18 GB] Using butterworth filter with order 8.
> [2026-06-09 15:17:39.21 GMT-4] [CPU: 350.1 MB] [Avail: 314.17 GB] ===== Masking =====
> [2026-06-09 15:17:41.74 GMT-4] [CPU: 882.3 MB] [Avail: 315.20 GB] No mask input was connected, so dynamic masking will be
enabled.
> [2026-06-09 15:17:41.75 GMT-4] [CPU: 882.3 MB] [Avail: 315.19 GB] Dynamic mask threshold: 0.2000
> [2026-06-09 15:17:41.75 GMT-4] [CPU: 882.3 MB] [Avail: 315.19 GB] Dynamic mask near (A): 6.00
> [2026-06-09 15:17:41.75 GMT-4] [CPU: 882.3 MB] [Avail: 315.18 GB] Dynamic mask far (A): 14.00
> [2026-06-09 15:17:41.76 GMT-4] [CPU: 882.3 MB] [Avail: 315.18 GB] ===== Initial Model =====
> [2026-06-09 15:17:41.76 GMT-4] [CPU: 882.3 MB] [Avail: 315.17 GB] Resampling initial model to specified volume representation
size and pixel-size...
> [2026-06-09 15:17:44.87 GMT-4] [CPU: 1.26 GB] [Avail: 314.09 GB] Estimating scale of initial reference.
> [2026-06-09 15:17:47.59 GMT-4] [CPU: 77.4 MB] [Avail: 313.29 GB] WARNING: io_uring support disabled (not supported by kernel),
I/O performance may degrade
> [2026-06-09 15:17:51.46 GMT-4] [CPU: 1.47 GB] [Avail: 314.05 GB] Rescaling initial reference by a factor of 1.448
> [2026-06-09 15:17:53.29 GMT-4] [CPU: 1.47 GB] [Avail: 313.74 GB] Estimating scale of initial reference.
> [2026-06-09 15:17:56.23 GMT-4] [CPU: 1.43 GB] [Avail: 311.64 GB] Rescaling initial reference by a factor of 0.998
> [2026-06-09 15:17:57.82 GMT-4] [CPU: 1.43 GB] [Avail: 310.31 GB] Estimating scale of initial reference.
> [2026-06-09 15:18:01.00 GMT-4] [CPU: 1.43 GB] [Avail: 313.29 GB] Rescaling initial reference by a factor of 1.003
> [2026-06-09 15:18:03.76 GMT-4] Initial Real Space Slices

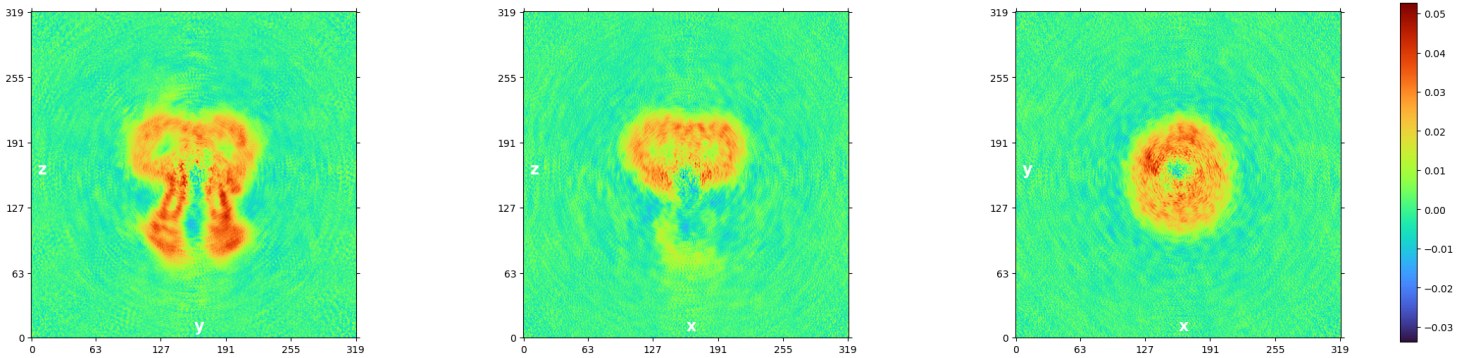
```



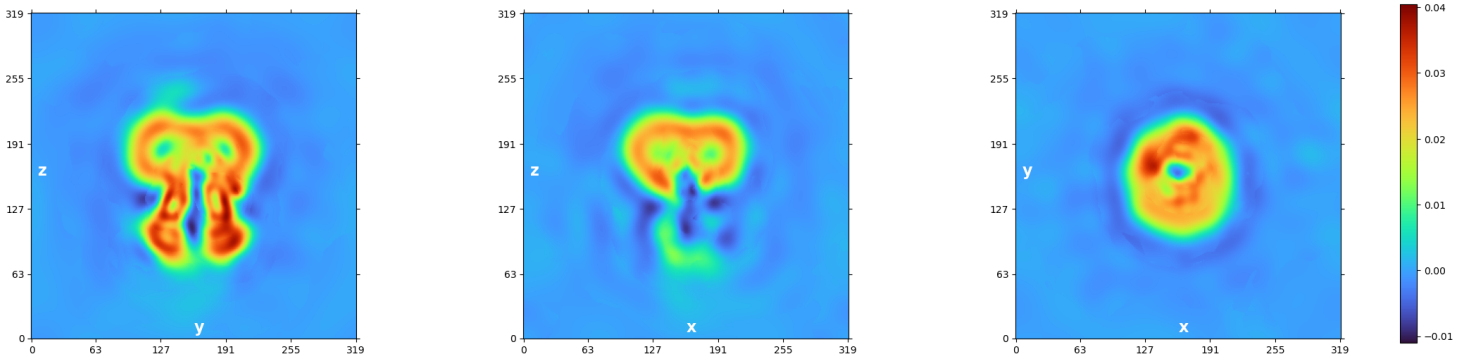
> [2026-06-09 15:18:05.28 GMT-4] Initial Fourier Space Slices



```
> [2026-06-09 15:18:05.28 GMT-4] [CPU: 1.49 GB] [Avail: 311.84 GB] ===== Starting Refinement Iterations =====
> [2026-06-09 15:18:05.28 GMT-4] [CPU: 1.49 GB] [Avail: 311.83 GB] ----- Start Iteration 0
> [2026-06-09 15:18:05.29 GMT-4] [CPU: 1.49 GB] [Avail: 311.82 GB] Using Max Alignment Radius 8.853 (30.000A)
> [2026-06-09 15:18:05.29 GMT-4] [CPU: 1.49 GB] [Avail: 311.81 GB] Auto batchsize: 5942 in each split
> [2026-06-09 15:18:09.31 GMT-4] [CPU: 2.00 GB] [Avail: 308.67 GB] -- THR 1 BATCH 500 NUM 1471 TOTAL 3.2919192 ELAPSED 31.617183
--
> [2026-06-09 15:18:43.25 GMT-4] [CPU: 3.19 GB] [Avail: 302.21 GB] Processed 11884.000 images in 33.939s.
> [2026-06-09 15:18:47.51 GMT-4] [CPU: 3.51 GB] [Avail: 301.53 GB] Computing FSCs...
> [2026-06-09 15:18:47.52 GMT-4] [CPU: 3.51 GB] [Avail: 301.50 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:18:47.53 GMT-4] [CPU: 3.51 GB] [Avail: 301.49 GB] Computing FFTs on GPU.
> [2026-06-09 15:18:50.03 GMT-4] [CPU: 3.76 GB] [Avail: 300.93 GB] Done in 2.508s
> [2026-06-09 15:18:50.03 GMT-4] [CPU: 3.76 GB] [Avail: 300.97 GB] Computing cFSCs...
> [2026-06-09 15:18:52.04 GMT-4] [CPU: 3.76 GB] [Avail: 301.90 GB] Done in 2.000s
> [2026-06-09 15:18:52.04 GMT-4] [CPU: 3.76 GB] [Avail: 301.90 GB] Using Filter Radius 29.366 (9.044A) | Previous: 8.853
(30.000A)
> [2026-06-09 15:18:58.21 GMT-4] [CPU: 4.83 GB] [Avail: 302.03 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:18:58.22 GMT-4] [CPU: 4.83 GB] [Avail: 302.03 GB] Running local cross validation for A ...
> [2026-06-09 15:19:31.23 GMT-4] [CPU: 7.51 GB] [Avail: 300.12 GB] Local cross validation A done in 33.011s
> [2026-06-09 15:19:32.77 GMT-4] FSC Filtered Side A
```

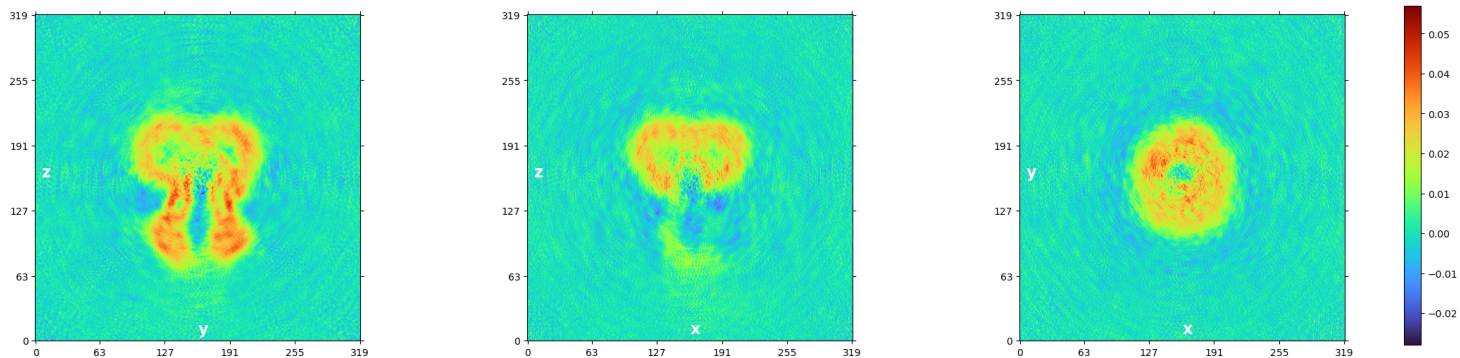


> [2026-06-09 15:19:33.99 GMT-4] CV Filtered Side A

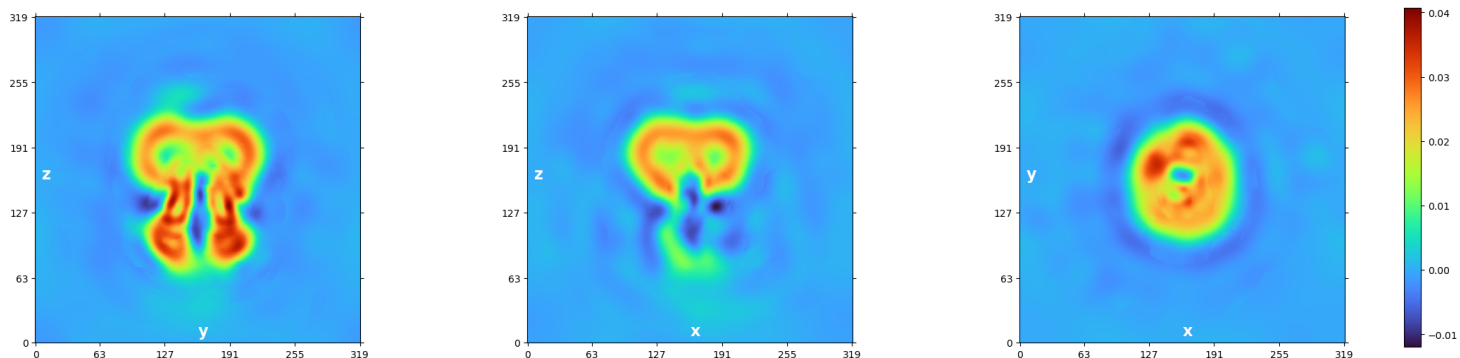


```
> [2026-06-09 15:19:33.99 GMT-4] [CPU: 7.51 GB] [Avail: 301.23 GB] Running local cross validation for B ...
> [2026-06-09 15:19:58.18 GMT-4] [CPU: 7.79 GB] [Avail: 297.31 GB] Local cross validation B done in 24.187s
```

> [2026-06-09 15:19:59.53 GMT-4] FSC Filtered Side B



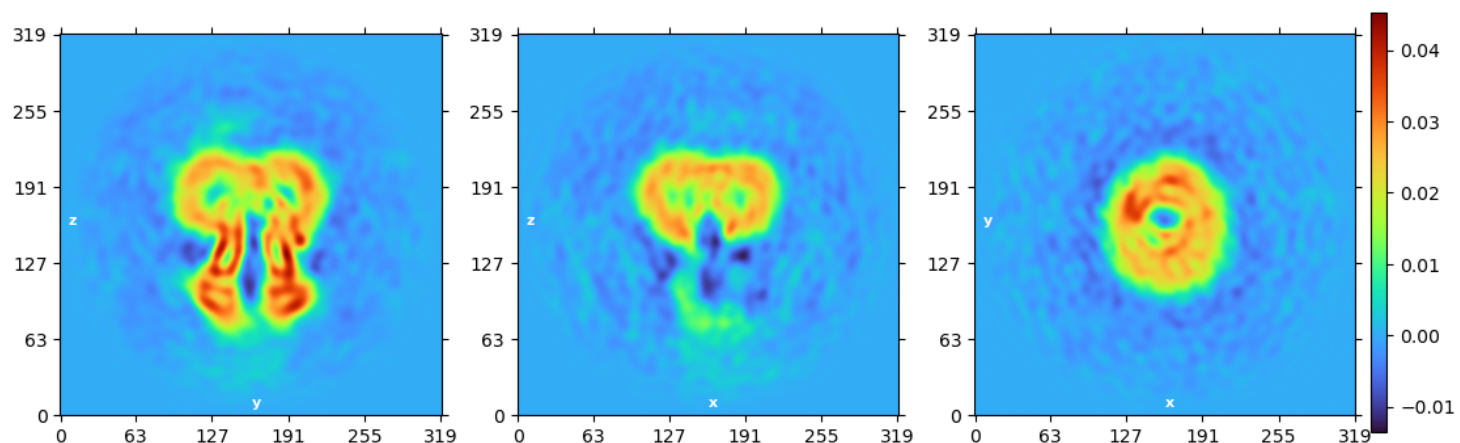
> [2026-06-09 15:20:00.57 GMT-4] CV Filtered Side B



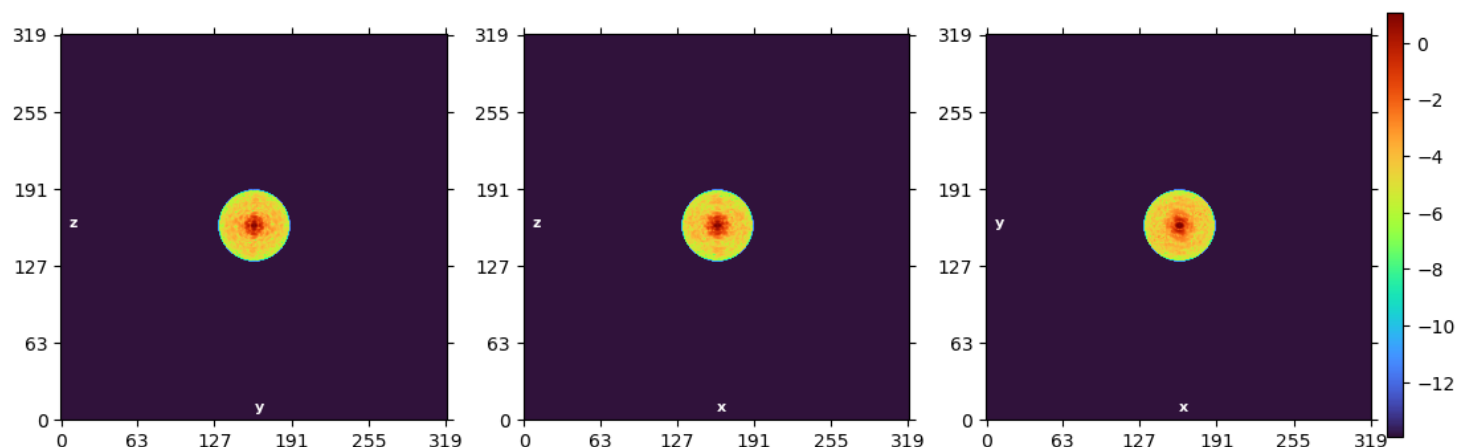
> [2026-06-09 15:20:08.96 GMT-4] [CPU: 8.12 GB] [Avail: 300.61 GB] Estimated Bfactor: -367.4

> [2026-06-09 15:20:08.97 GMT-4] [CPU: 8.12 GB] [Avail: 300.65 GB] Plotting..

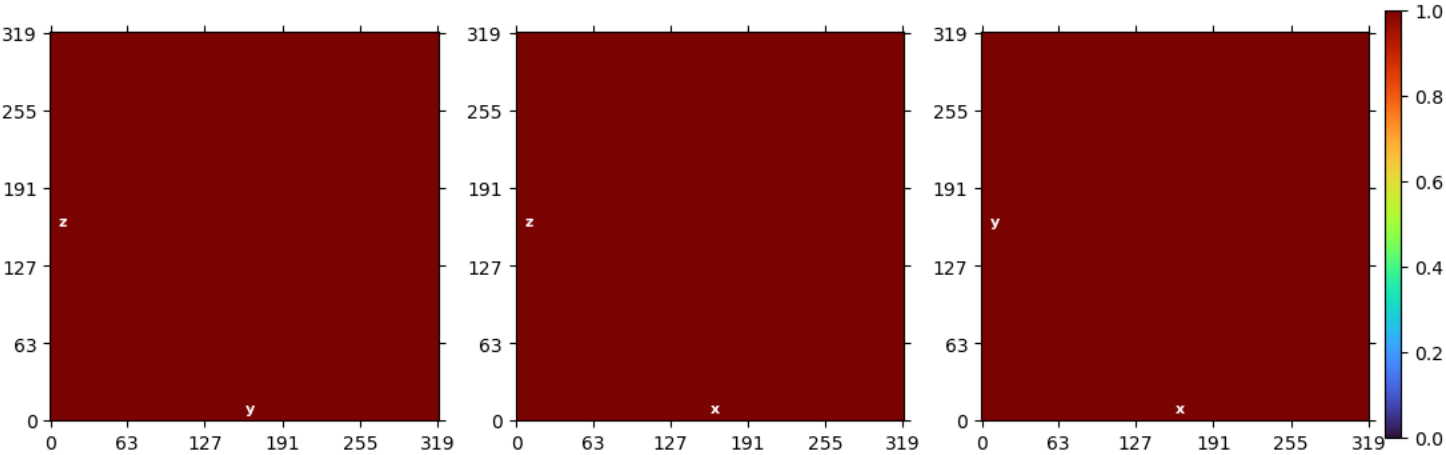
> [2026-06-09 15:20:15.46 GMT-4] Real Space Slices Iteration 000



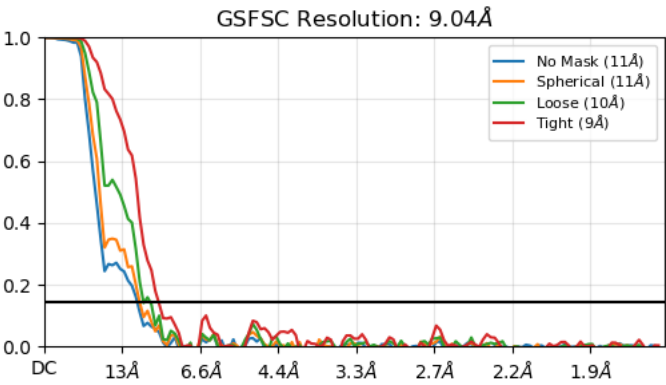
> [2026-06-09 15:20:16.62 GMT-4] Fourier Space Slices Iteration 000



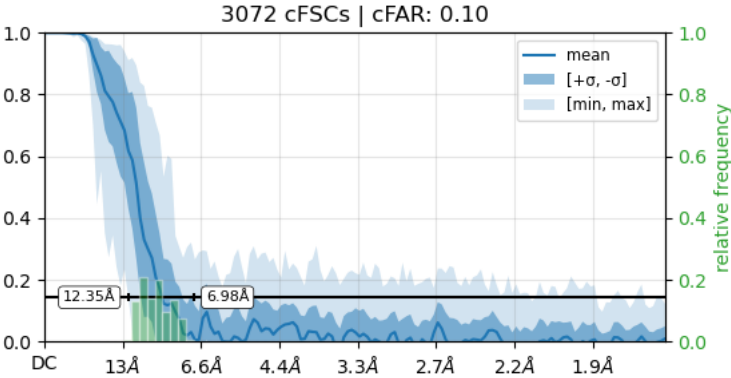
> [2026-06-09 15:20:17.50 GMT-4] Real Space Mask Slices Iteration 000



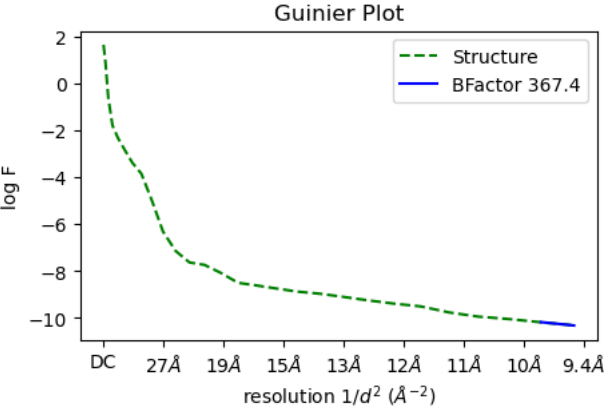
> [2026-06-09 15:20:18.01 GMT-4] FSC Iteration 000



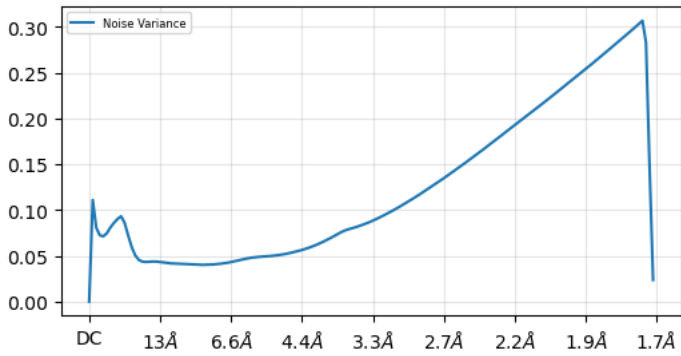
> [2026-06-09 15:20:18.69 GMT-4] cFSCs (Half-angle: 20°) Iteration 000, with tight mask



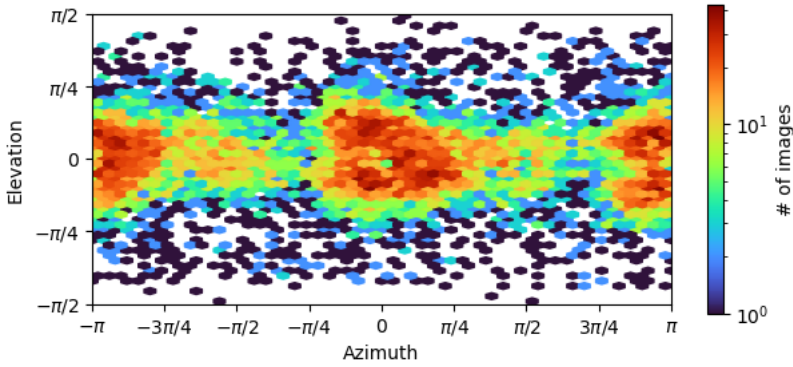
> [2026-06-09 15:20:24.22 GMT-4] Guinier Plot Iteration 000



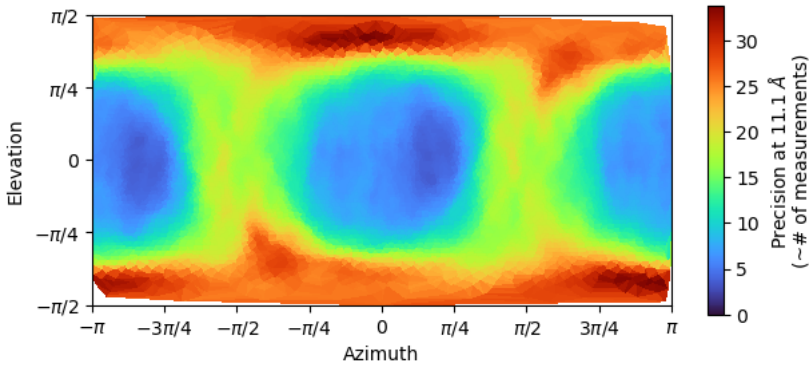
> [2026-06-09 15:20:24.54 GMT-4] Noise Model Iteration 000



> [2026-06-09 15:20:25.49 GMT-4] Viewing Direction Distribution Iteration 000

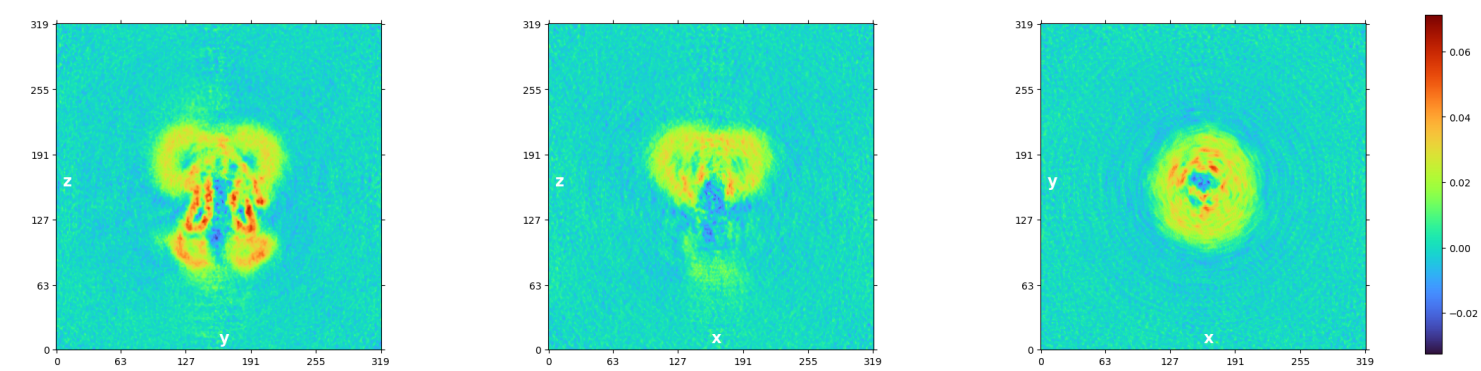


> [2026-06-09 15:20:26.87 GMT-4] Posterior Precision Directional Distribution Iteration 000

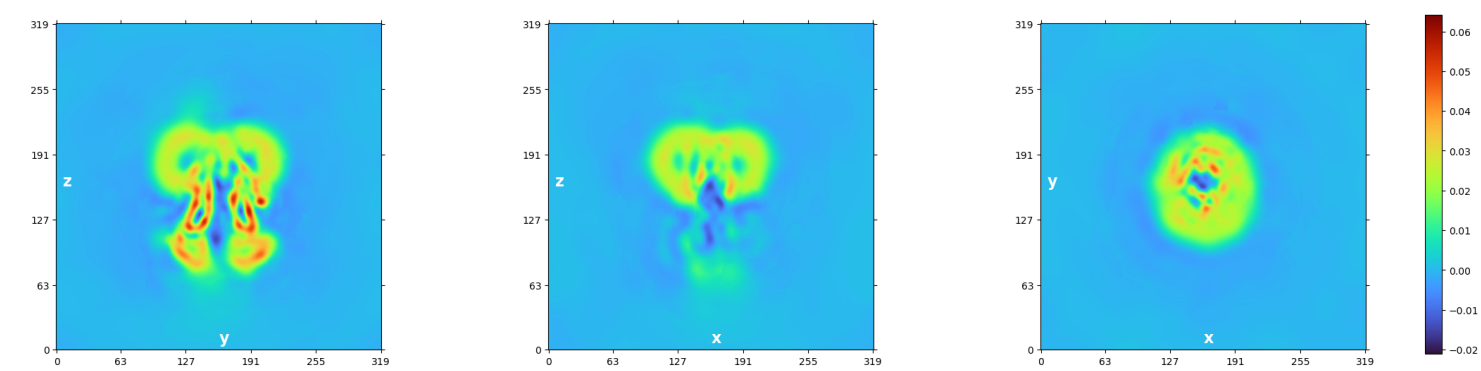


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> [2026-06-09 15:20:26.87 GMT-4] [CPU: 8.54 GB] [Avail: 298.70 GB] Done in 17.901s.
> [2026-06-09 15:20:26.87 GMT-4] [CPU: 8.54 GB] [Avail: 298.70 GB] Outputting files..
> [2026-06-09 15:20:41.28 GMT-4] [CPU: 8.98 GB] [Avail: 295.92 GB] Done in 14.404s.
> [2026-06-09 15:20:41.28 GMT-4] [CPU: 8.98 GB] [Avail: 295.91 GB] Done iteration 0 in 155.997s. Total time so far 190.495s
> [2026-06-09 15:20:41.99 GMT-4] [CPU: 8.98 GB] [Avail: 294.75 GB] ----- Start Iteration 1
> [2026-06-09 15:20:42.00 GMT-4] [CPU: 8.98 GB] [Avail: 294.75 GB] Using Max Alignment Radius 29.366 (9.044A)
> [2026-06-09 15:20:42.00 GMT-4] [CPU: 8.98 GB] [Avail: 294.75 GB] Auto batchsize: 7972 in each split
> [2026-06-09 15:20:42.00 GMT-4] [CPU: 8.98 GB] [Avail: 294.74 GB] Using dynamic mask.
> [2026-06-09 15:21:12.77 GMT-4] [CPU: 9.47 GB] [Avail: 297.83 GB] -- THR 0 BATCH 500 NUM 1986 TOTAL 4.3850893 ELAPSED 36.487824
--
> [2026-06-09 15:21:51.56 GMT-4] [CPU: 10.33 GB] [Avail: 291.09 GB] Processed 15925.000 images in 38.793s.
> [2026-06-09 15:21:54.85 GMT-4] [CPU: 10.37 GB] [Avail: 293.80 GB] Computing FSCs...
> [2026-06-09 15:21:54.86 GMT-4] [CPU: 10.37 GB] [Avail: 293.85 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:21:54.86 GMT-4] [CPU: 10.37 GB] [Avail: 293.84 GB] Computing FFTs on GPU.
> [2026-06-09 15:21:56.91 GMT-4] [CPU: 10.37 GB] [Avail: 295.84 GB] Done in 2.056s
> [2026-06-09 15:21:56.92 GMT-4] [CPU: 10.37 GB] [Avail: 295.83 GB] Computing cFSCs...
> [2026-06-09 15:21:58.84 GMT-4] [CPU: 10.37 GB] [Avail: 293.83 GB] Done in 1.917s
> [2026-06-09 15:21:58.84 GMT-4] [CPU: 10.37 GB] [Avail: 293.87 GB] Using Filter Radius 41.231 (6.442A) | Previous: 29.366
(9.044A)
> [2026-06-09 15:22:05.20 GMT-4] [CPU: 9.74 GB] [Avail: 294.77 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:22:05.21 GMT-4] [CPU: 9.74 GB] [Avail: 294.81 GB] Running local cross validation for A ...
> [2026-06-09 15:22:27.72 GMT-4] [CPU: 9.83 GB] [Avail: 294.56 GB] Local cross validation A done in 22.483s
```

> [2026-06-09 15:22:29.25 GMT-4] FSC Filtered Side A



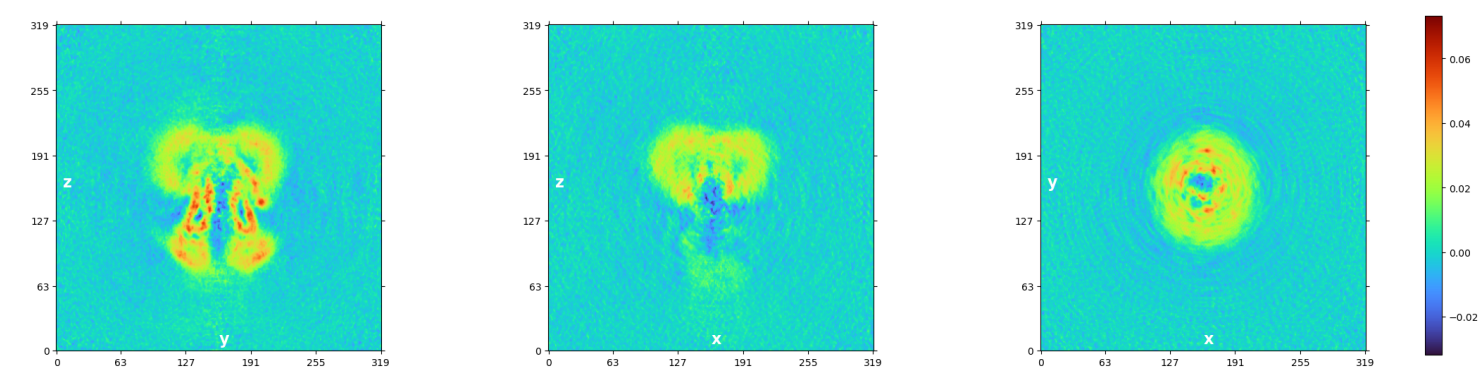
> [2026-06-09 15:22:30.22 GMT-4] CV Filtered Side A



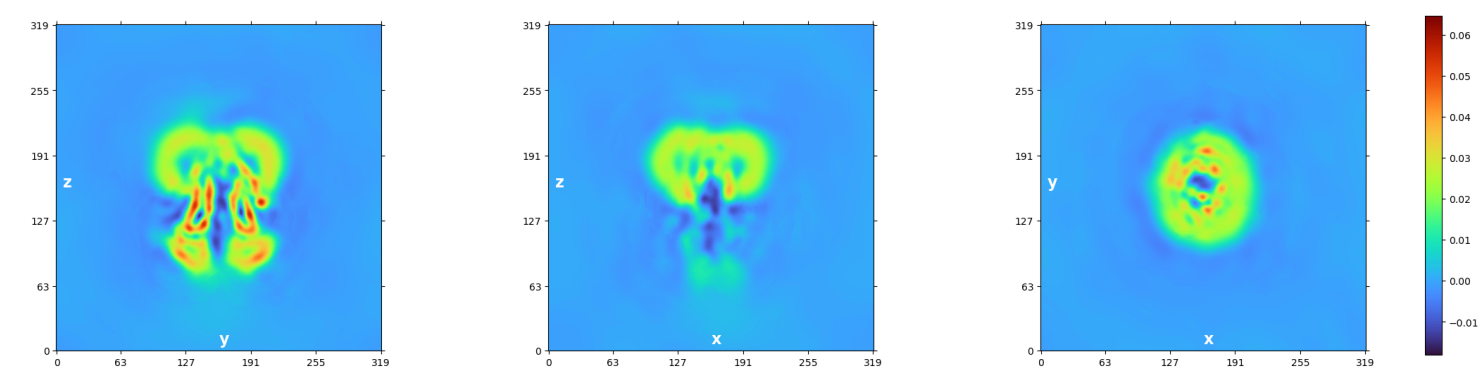
> [2026-06-09 15:22:30.23 GMT-4] [CPU: 9.83 GB] [Avail: 293.63 GB] Running local cross validation for B ...

> [2026-06-09 15:22:54.05 GMT-4] [CPU: 10.08 GB] [Avail: 294.09 GB] Local cross validation B done in 23.823s

> [2026-06-09 15:22:55.62 GMT-4] FSC Filtered Side B



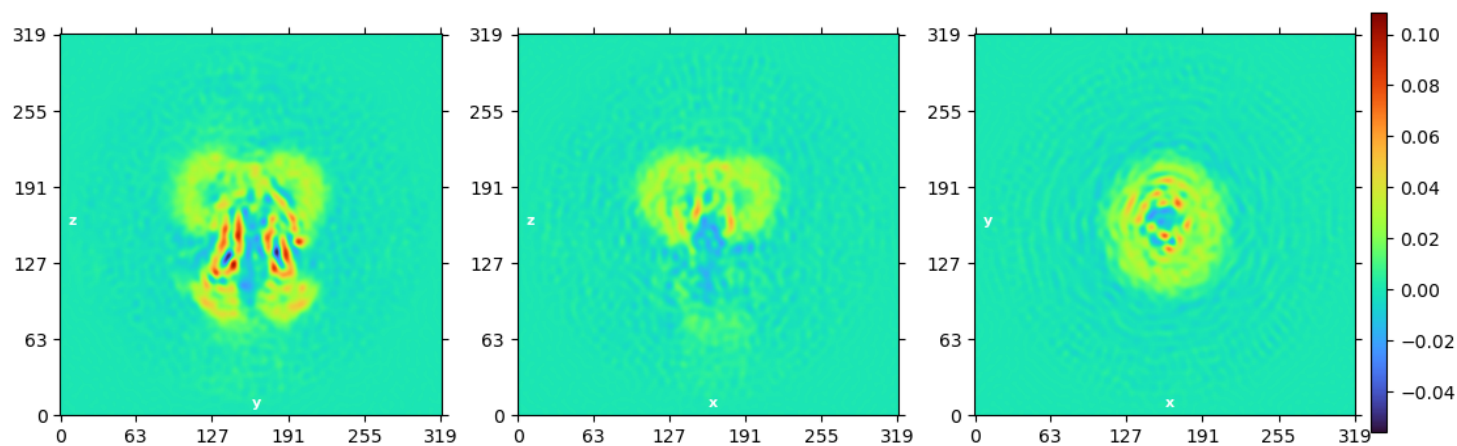
> [2026-06-09 15:22:56.67 GMT-4] CV Filtered Side B



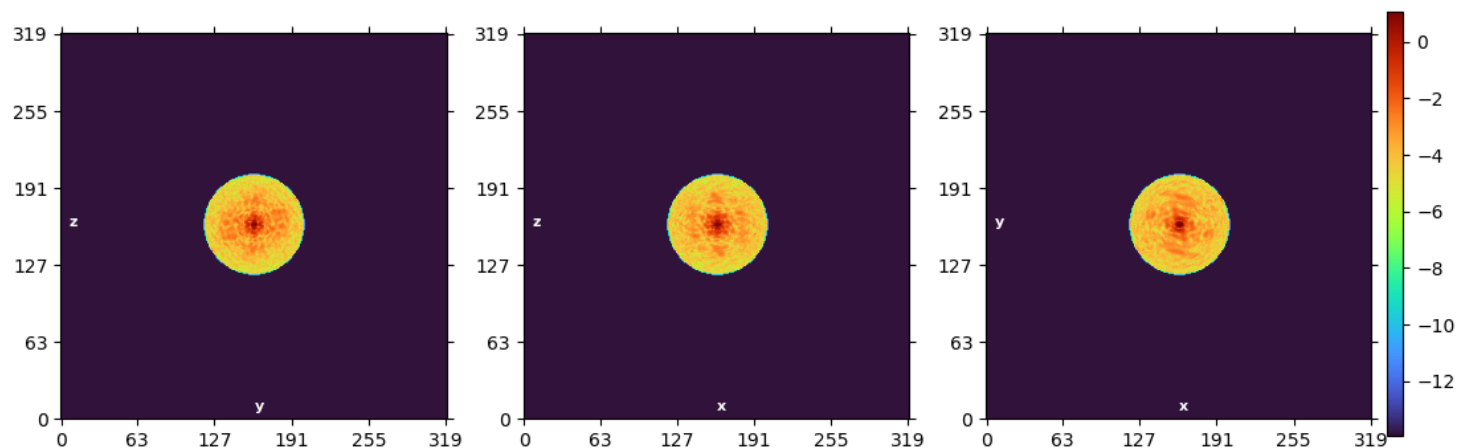
> [2026-06-09 15:23:06.27 GMT-4] [CPU: 10.30 GB] [Avail: 293.80 GB] Estimated Bfactor: -392.4

> [2026-06-09 15:23:06.28 GMT-4] [CPU: 10.30 GB] [Avail: 293.75 GB] Plotting..

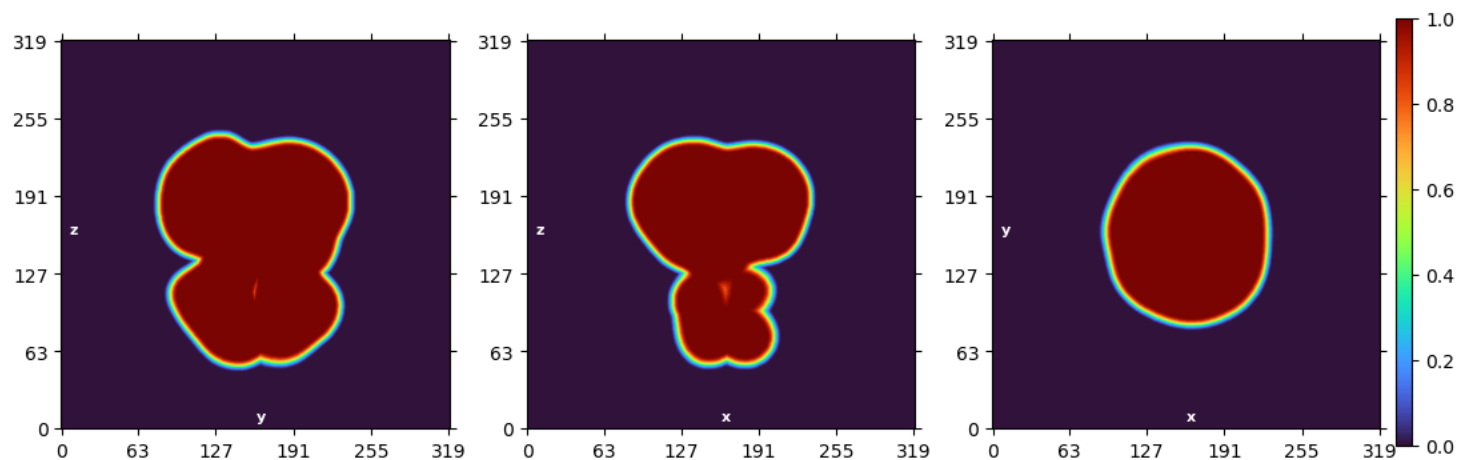
> [2026-06-09 15:23:13.21 GMT-4] Real Space Slices Iteration 001



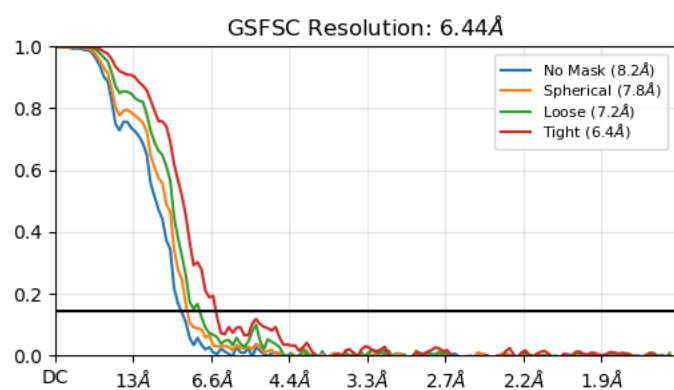
> [2026-06-09 15:23:14.58 GMT-4] Fourier Space Slices Iteration 001



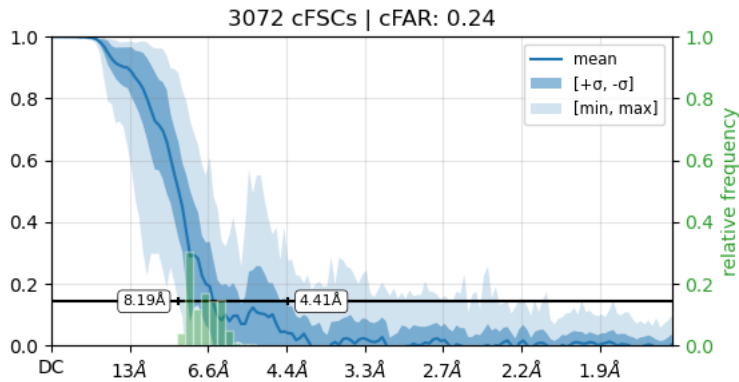
> [2026-06-09 15:23:16.57 GMT-4] Real Space Mask Slices Iteration 001



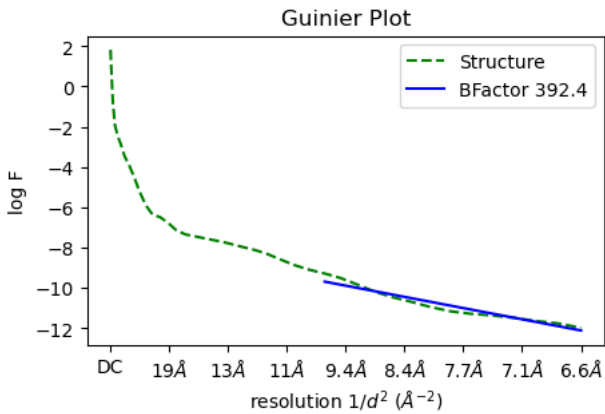
> [2026-06-09 15:23:17.12 GMT-4] FSC Iteration 001



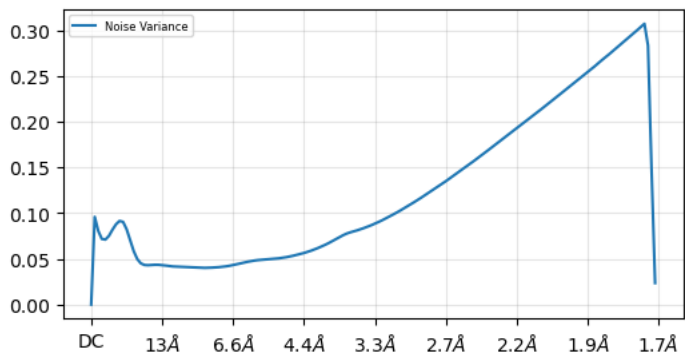
> [2026-06-09 15:23:18.04 GMT-4] cFSCs (Half-angle: 20°) Iteration 001, with tight mask



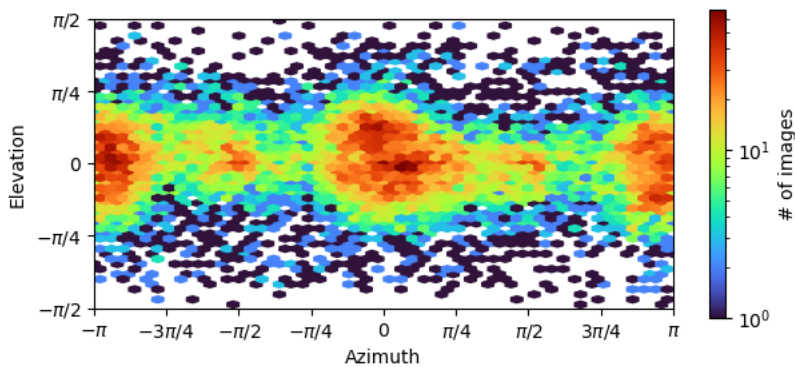
> [2026-06-09 15:23:23.73 GMT-4] Guinier Plot Iteration 001



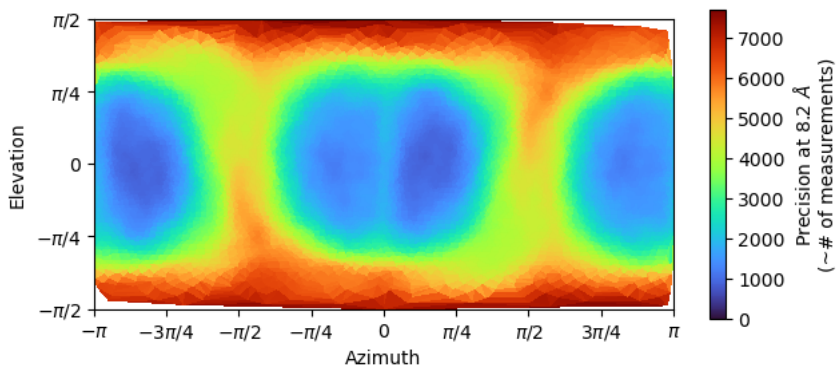
> [2026-06-09 15:23:24.04 GMT-4] Noise Model Iteration 001



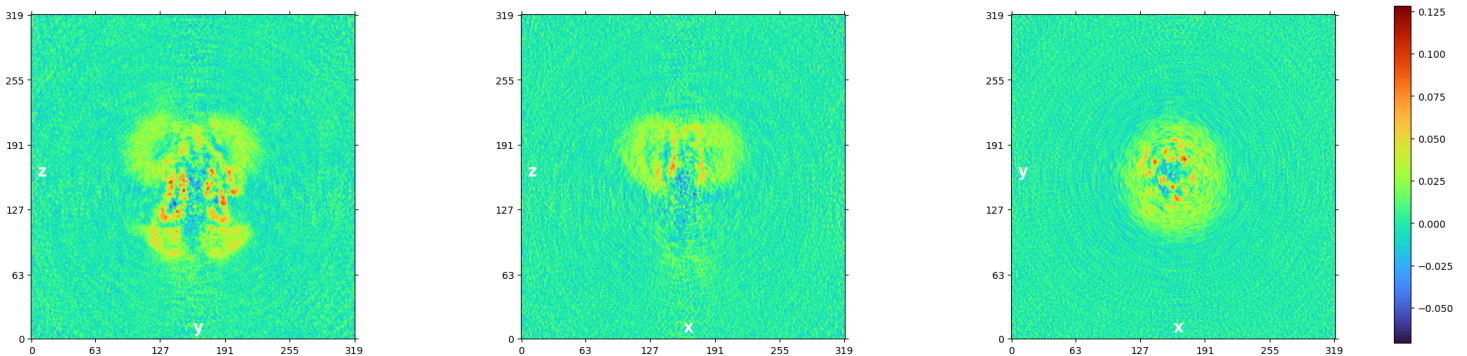
> [2026-06-09 15:23:24.80 GMT-4] Viewing Direction Distribution Iteration 001



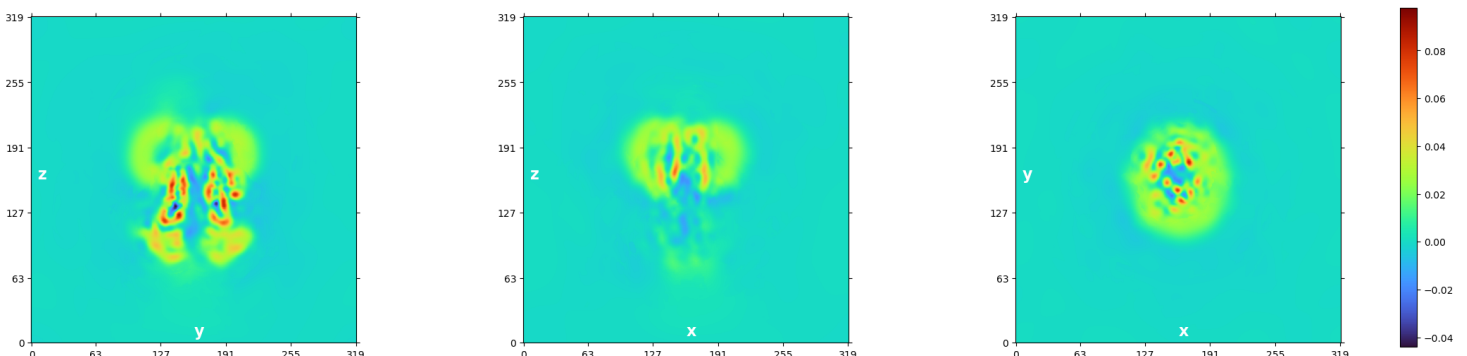
> [2026-06-09 15:23:26.13 GMT-4] Posterior Precision Directional Distribution Iteration 001



```
> [2026-06-09 15:23:26.14 GMT-4] [CPU: 10.34 GB] [Avail: 293.35 GB] Done in 19.855s.
> [2026-06-09 15:23:26.14 GMT-4] [CPU: 10.34 GB] [Avail: 293.34 GB] Outputting files..
> [2026-06-09 15:23:41.47 GMT-4] [CPU: 10.24 GB] [Avail: 294.67 GB] Done in 15.326s.
> [2026-06-09 15:23:41.47 GMT-4] [CPU: 10.24 GB] [Avail: 294.66 GB] Done iteration 1 in 179.475s. Total time so far 370.682s
> [2026-06-09 15:23:42.38 GMT-4] [CPU: 10.24 GB] [Avail: 293.14 GB] ----- Start Iteration 2
> [2026-06-09 15:23:42.38 GMT-4] [CPU: 10.24 GB] [Avail: 293.13 GB] Using Max Alignment Radius 41.231 (6.442Å)
> [2026-06-09 15:23:42.39 GMT-4] [CPU: 10.24 GB] [Avail: 293.13 GB] Auto batchsize: 7972 in each split
> [2026-06-09 15:23:42.39 GMT-4] [CPU: 10.24 GB] [Avail: 293.13 GB] Using dynamic mask.
> [2026-06-09 15:24:12.37 GMT-4] [CPU: 10.24 GB] [Avail: 293.05 GB] -- THR 0 BATCH 500 NUM 2000 TOTAL 5.3673038 ELAPSED 35.842704
--
> [2026-06-09 15:24:50.51 GMT-4] [CPU: 11.10 GB] [Avail: 294.40 GB] Processed 15925.000 images in 38.146s.
> [2026-06-09 15:24:53.64 GMT-4] [CPU: 11.13 GB] [Avail: 291.95 GB] Computing FSCs...
> [2026-06-09 15:24:53.64 GMT-4] [CPU: 11.13 GB] [Avail: 291.95 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:24:53.65 GMT-4] [CPU: 11.13 GB] [Avail: 291.95 GB] Computing FFTs on GPU.
> [2026-06-09 15:24:55.58 GMT-4] [CPU: 11.13 GB] [Avail: 291.73 GB] Done in 1.941s
> [2026-06-09 15:24:55.59 GMT-4] [CPU: 11.13 GB] [Avail: 291.73 GB] Computing cFSCs...
> [2026-06-09 15:24:57.56 GMT-4] [CPU: 11.13 GB] [Avail: 291.16 GB] Done in 1.971s
> [2026-06-09 15:24:57.57 GMT-4] [CPU: 11.13 GB] [Avail: 291.16 GB] Using Filter Radius 63.788 (4.164Å) | Previous: 41.231
(6.442Å)
> [2026-06-09 15:25:03.47 GMT-4] [CPU: 10.38 GB] [Avail: 292.94 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:25:03.47 GMT-4] [CPU: 10.38 GB] [Avail: 292.94 GB] Running local cross validation for A ...
> [2026-06-09 15:25:25.44 GMT-4] [CPU: 10.60 GB] [Avail: 291.29 GB] Local cross validation A done in 21.971s
> [2026-06-09 15:25:26.78 GMT-4] FSC Filtered Side A
```



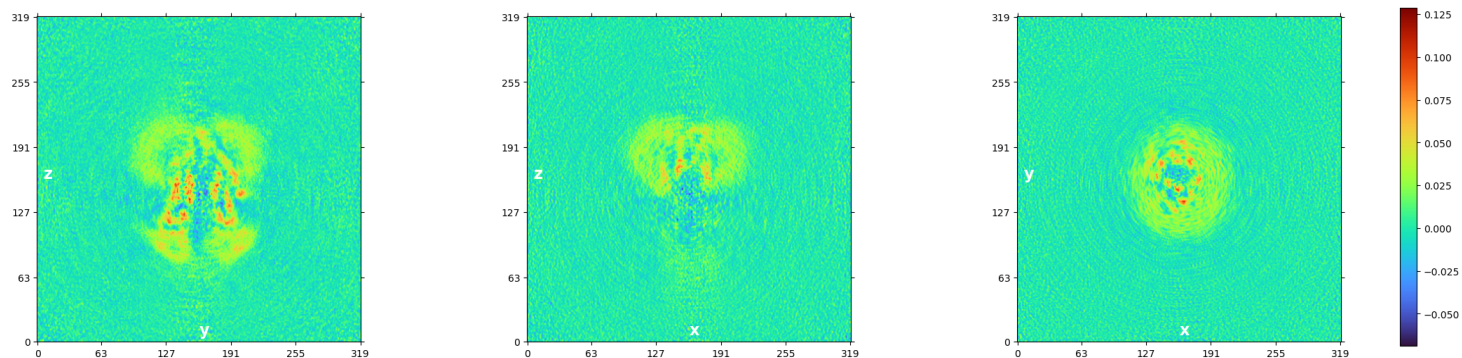
> [2026-06-09 15:25:27.77 GMT-4] CV Filtered Side A



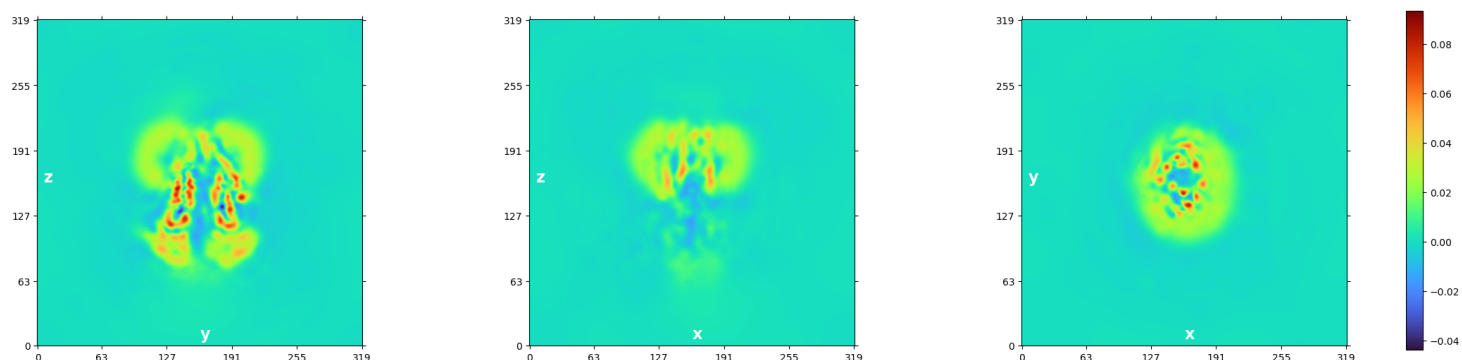
```
> [2026-06-09 15:25:27.77 GMT-4] [CPU: 10.60 GB] [Avail: 291.27 GB] Running local cross validation for B ...
```

> [2026-06-09 15:25:50.19 GMT-4] [CPU: 10.85 GB] [Avail: 293.00 GB] Local cross validation B done in 22.417s

> [2026-06-09 15:25:51.70 GMT-4] FSC Filtered Side B



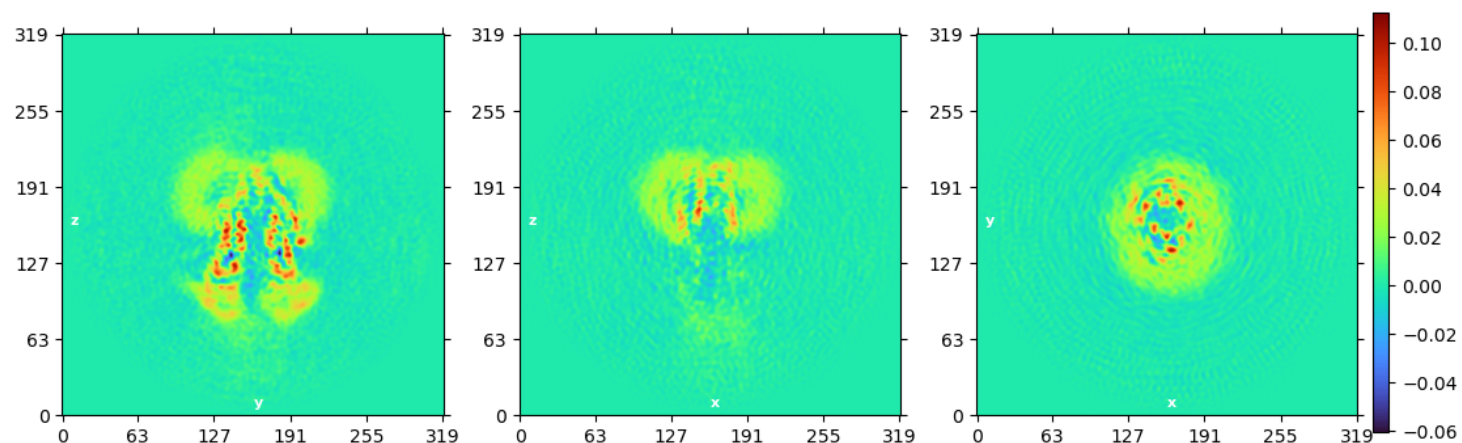
> [2026-06-09 15:25:52.94 GMT-4] CV Filtered Side B



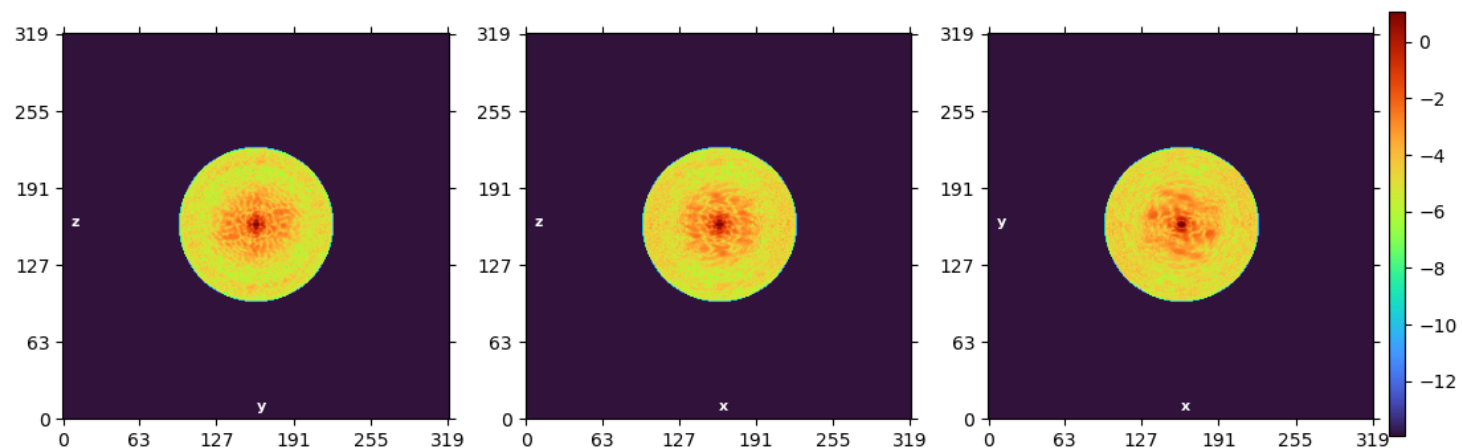
> [2026-06-09 15:26:03.13 GMT-4] [CPU: 10.87 GB] [Avail: 291.05 GB] Estimated Bfactor: -106.9

> [2026-06-09 15:26:03.14 GMT-4] [CPU: 10.87 GB] [Avail: 291.05 GB] Plotting..

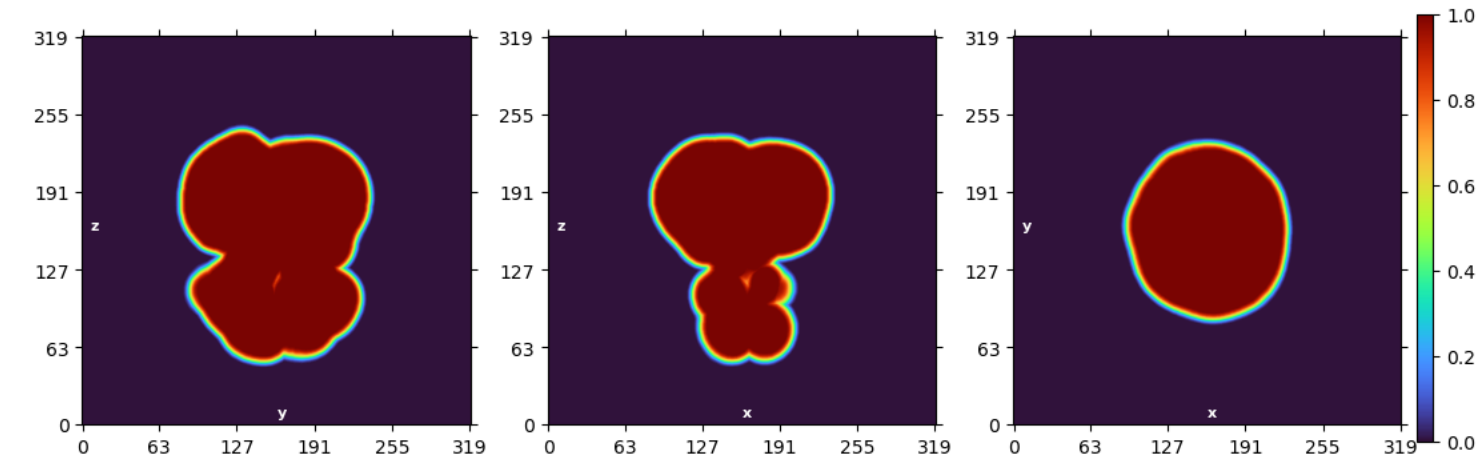
> [2026-06-09 15:26:10.71 GMT-4] Real Space Slices Iteration 002



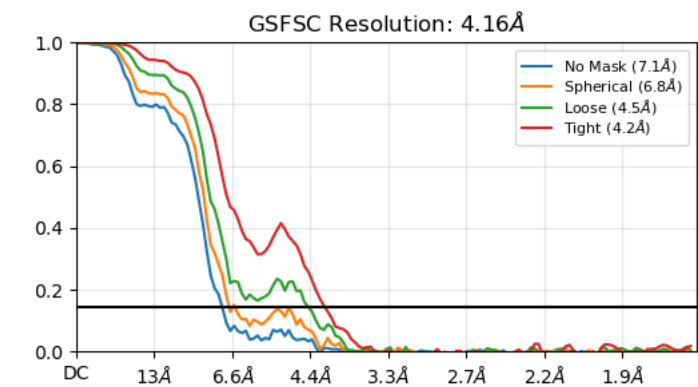
> [2026-06-09 15:26:12.15 GMT-4] Fourier Space Slices Iteration 002



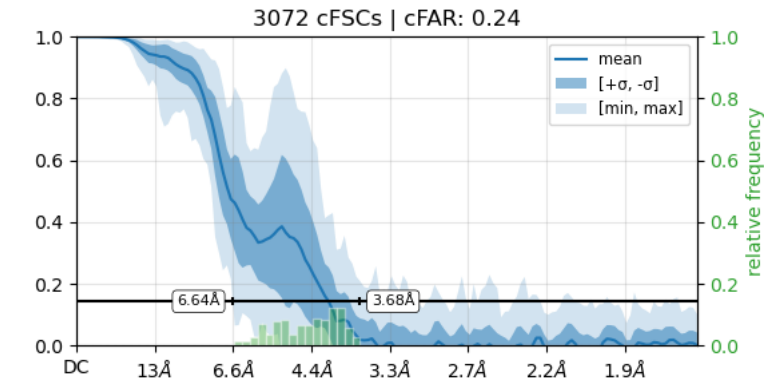
> [2026-06-09 15:26:14.33 GMT-4] Real Space Mask Slices Iteration 002



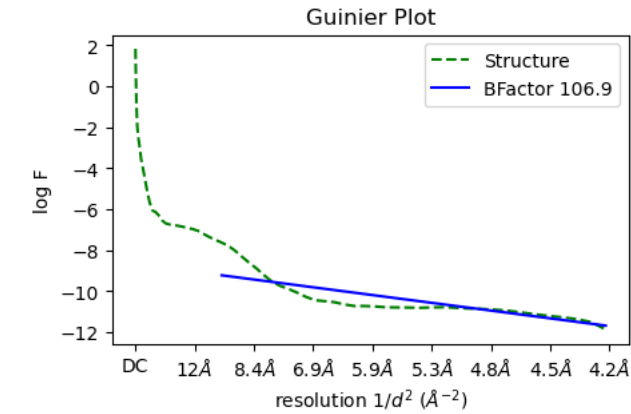
> [2026-06-09 15:26:14.90 GMT-4] FSC Iteration 002



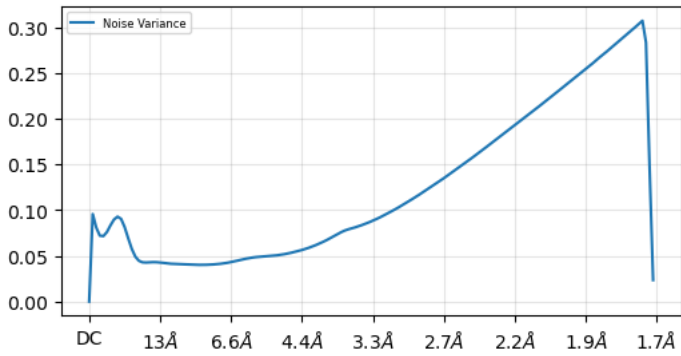
> [2026-06-09 15:26:15.76 GMT-4] cFSCs (Half-angle: 20°) Iteration 002, with tight mask



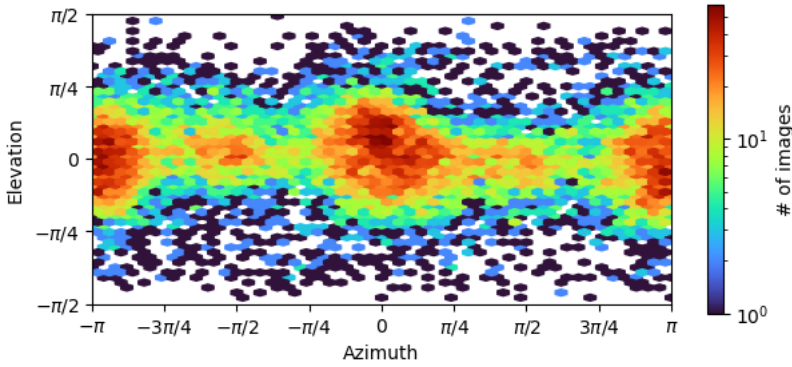
> [2026-06-09 15:26:21.49 GMT-4] Guinier Plot Iteration 002



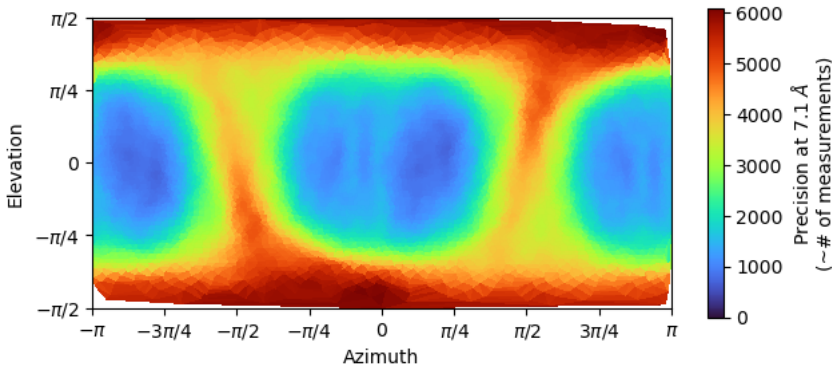
> [2026-06-09 15:26:21.82 GMT-4] Noise Model Iteration 002



> [2026-06-09 15:26:22.60 GMT-4] Viewing Direction Distribution Iteration 002

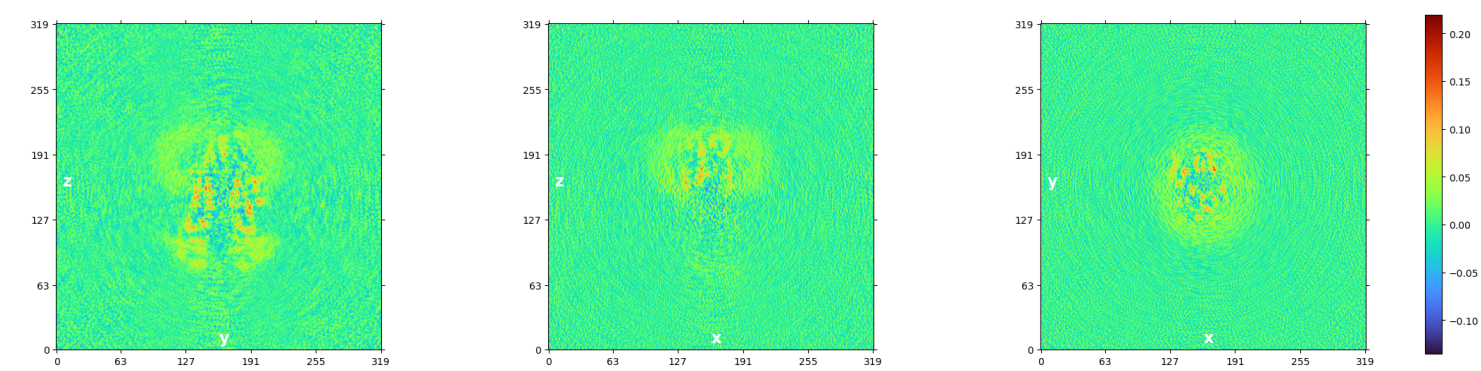


> [2026-06-09 15:26:24.19 GMT-4] Posterior Precision Directional Distribution Iteration 002

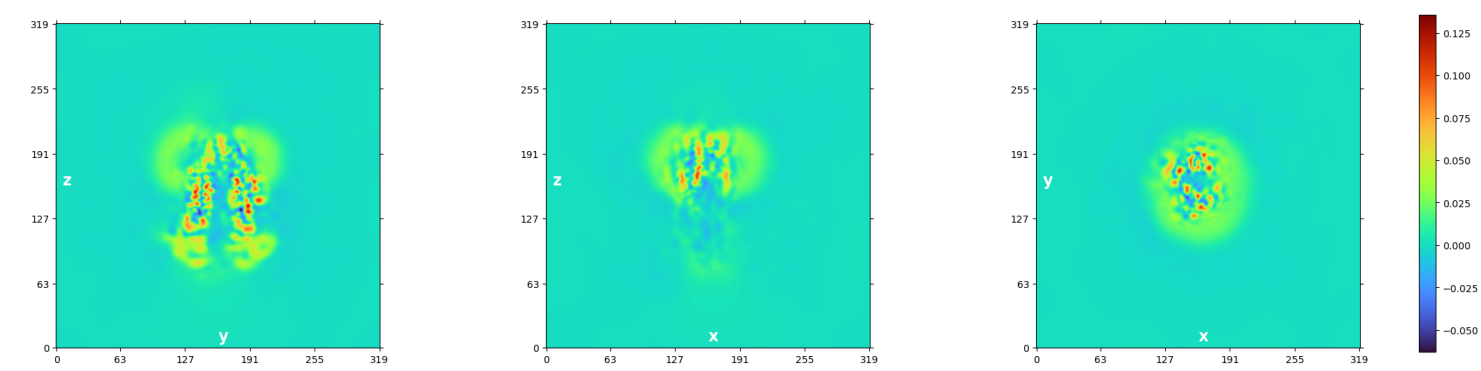


```
> [2026-06-09 15:26:24.19 GMT-4] [CPU: 10.88 GB] [Avail: 293.46 GB] Done in 21.056s.
> [2026-06-09 15:26:24.20 GMT-4] [CPU: 10.88 GB] [Avail: 293.46 GB] Outputting files..
> [2026-06-09 15:26:40.69 GMT-4] [CPU: 10.78 GB] [Avail: 291.27 GB] Done in 16.489s.
> [2026-06-09 15:26:40.69 GMT-4] [CPU: 10.78 GB] [Avail: 291.27 GB] Done iteration 2 in 178.311s. Total time so far 549.904s
> [2026-06-09 15:26:41.48 GMT-4] [CPU: 10.78 GB] [Avail: 292.62 GB] ----- Start Iteration 3
> [2026-06-09 15:26:41.49 GMT-4] [CPU: 10.78 GB] [Avail: 292.66 GB] Using Max Alignment Radius 63.788 (4.164A)
> [2026-06-09 15:26:41.49 GMT-4] [CPU: 10.78 GB] [Avail: 292.68 GB] Auto batchsize: 7972 in each split
> [2026-06-09 15:26:41.49 GMT-4] [CPU: 10.78 GB] [Avail: 292.72 GB] Using dynamic mask.
> [2026-06-09 15:27:11.54 GMT-4] [CPU: 10.78 GB] [Avail: 290.57 GB] -- THR 0 BATCH 500 NUM 1986 TOTAL 4.6459174 ELAPSED 37.314549
--
> [2026-06-09 15:27:51.17 GMT-4] [CPU: 11.54 GB] [Avail: 290.70 GB] Processed 15925.000 images in 39.633s.
> [2026-06-09 15:27:54.18 GMT-4] [CPU: 11.54 GB] [Avail: 289.69 GB] Computing FSCs...
> [2026-06-09 15:27:54.19 GMT-4] [CPU: 11.54 GB] [Avail: 289.73 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:27:54.19 GMT-4] [CPU: 11.54 GB] [Avail: 289.76 GB] Computing FFTs on GPU.
> [2026-06-09 15:27:56.08 GMT-4] [CPU: 11.54 GB] [Avail: 291.98 GB] Done in 1.894s
> [2026-06-09 15:27:56.08 GMT-4] [CPU: 11.54 GB] [Avail: 291.98 GB] Computing cFSCs...
> [2026-06-09 15:27:58.11 GMT-4] [CPU: 11.54 GB] [Avail: 291.31 GB] Done in 2.026s
> [2026-06-09 15:27:58.12 GMT-4] [CPU: 11.54 GB] [Avail: 291.28 GB] Using Filter Radius 68.698 (3.866A) | Previous: 63.788
(4.164A)
> [2026-06-09 15:28:03.97 GMT-4] [CPU: 10.78 GB] [Avail: 289.70 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:28:03.98 GMT-4] [CPU: 10.78 GB] [Avail: 289.70 GB] Running local cross validation for A ...
> [2026-06-09 15:28:27.56 GMT-4] [CPU: 11.04 GB] [Avail: 291.62 GB] Local cross validation A done in 23.573s
```

> [2026-06-09 15:28:28.94 GMT-4] FSC Filtered Side A



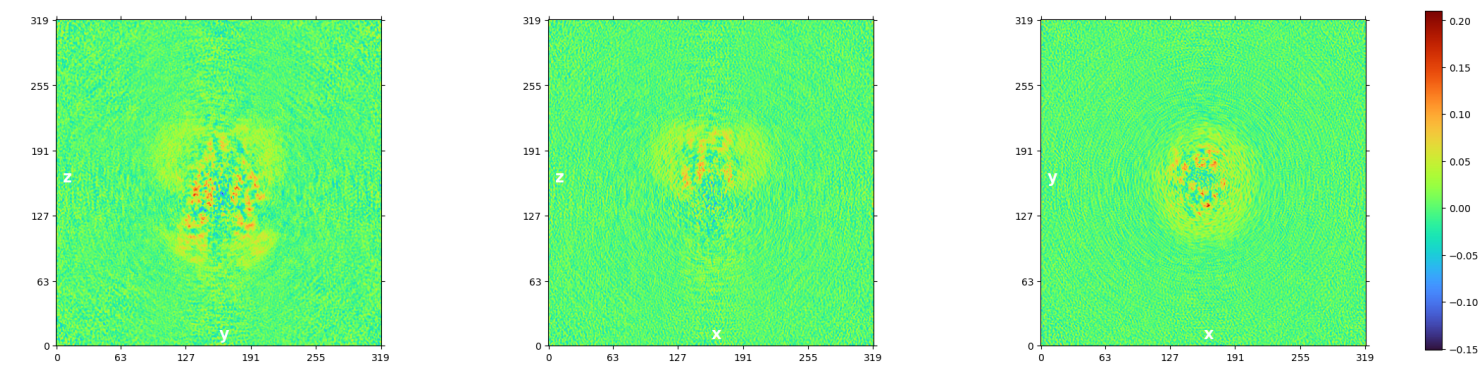
> [2026-06-09 15:28:30.07 GMT-4] CV Filtered Side A



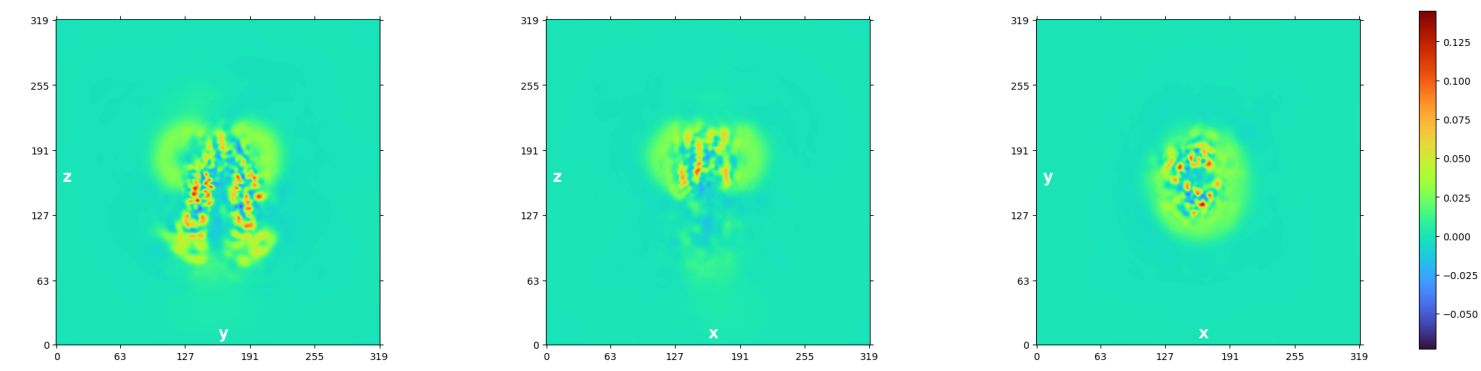
> [2026-06-09 15:28:30.07 GMT-4] [CPU: 11.04 GB] [Avail: 291.57 GB] Running local cross validation for B ...

> [2026-06-09 15:28:52.73 GMT-4] [CPU: 11.29 GB] [Avail: 300.63 GB] Local cross validation B done in 22.653s

> [2026-06-09 15:28:53.99 GMT-4] FSC Filtered Side B



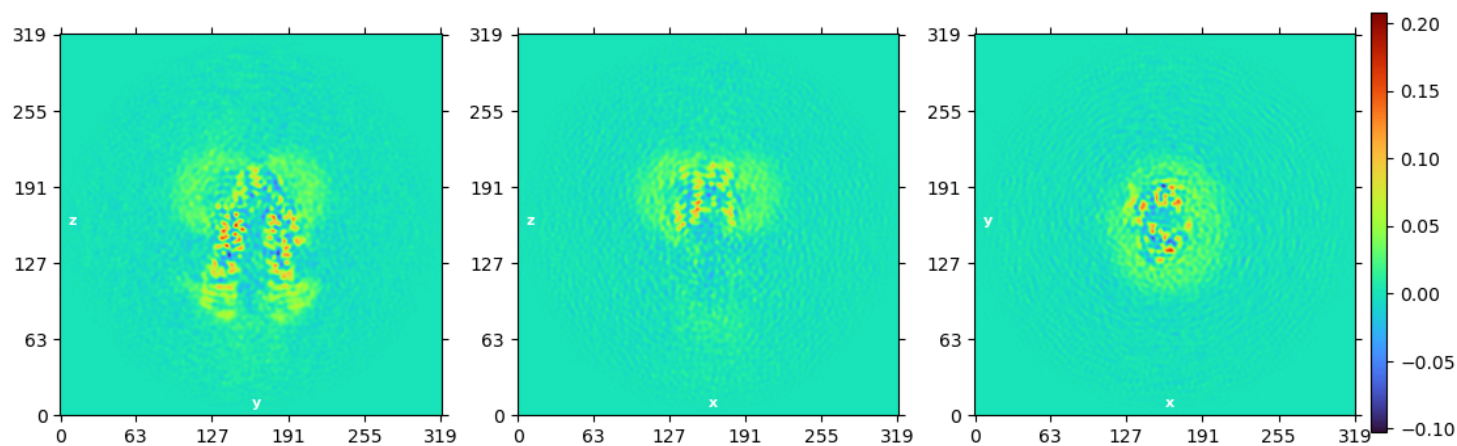
> [2026-06-09 15:28:54.97 GMT-4] CV Filtered Side B



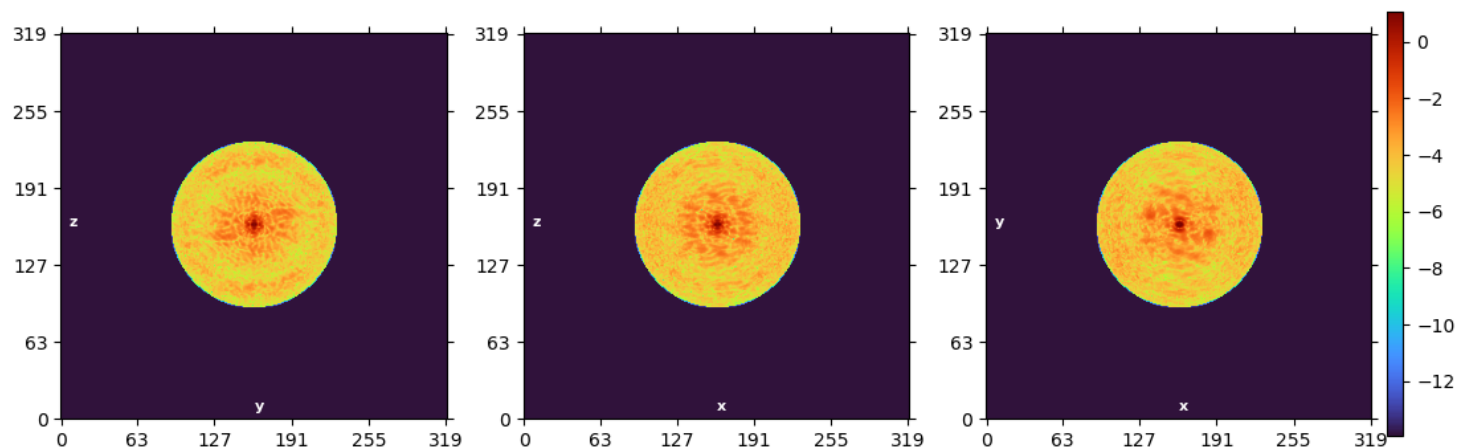
> [2026-06-09 15:29:05.59 GMT-4] [CPU: 11.31 GB] [Avail: 301.85 GB] Estimated Bfactor: -86.8

> [2026-06-09 15:29:05.60 GMT-4] [CPU: 11.31 GB] [Avail: 301.92 GB] Plotting..

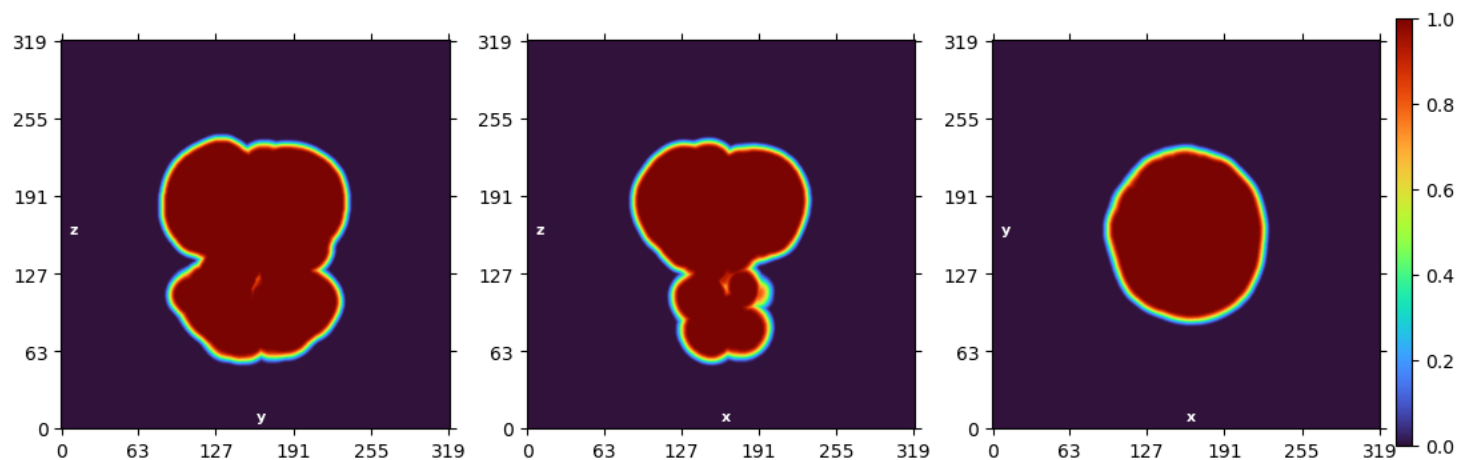
> [2026-06-09 15:29:12.46 GMT-4] Real Space Slices Iteration 003



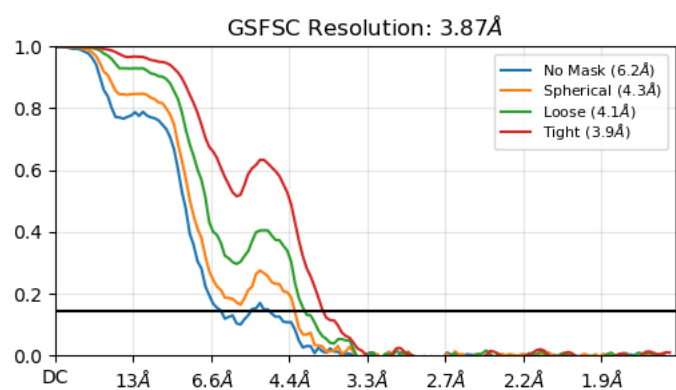
> [2026-06-09 15:29:13.64 GMT-4] Fourier Space Slices Iteration 003



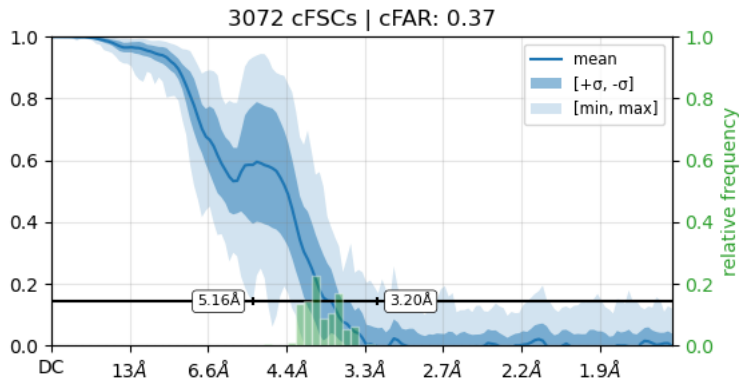
> [2026-06-09 15:29:15.55 GMT-4] Real Space Mask Slices Iteration 003



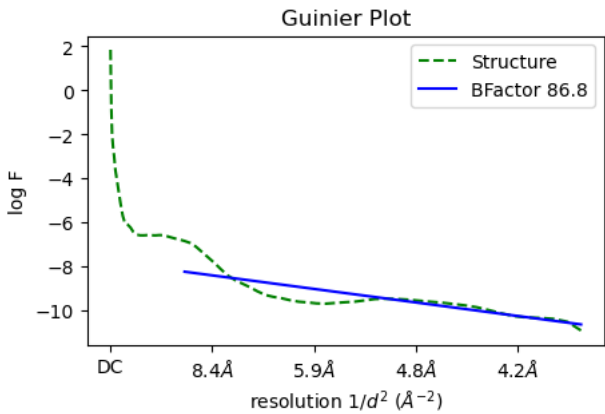
> [2026-06-09 15:29:16.14 GMT-4] FSC Iteration 003



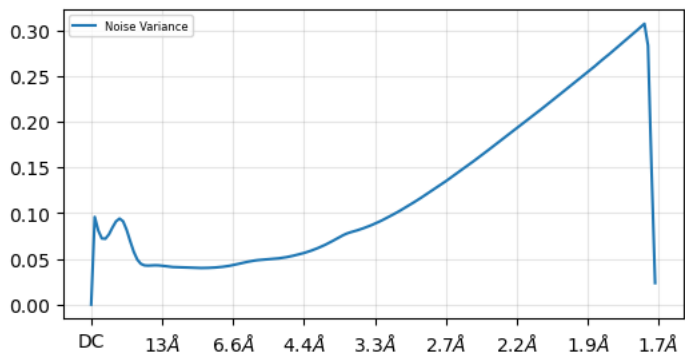
> [2026-06-09 15:29:17.03 GMT-4] cFSCs (Half-angle: 20°) Iteration 003, with tight mask



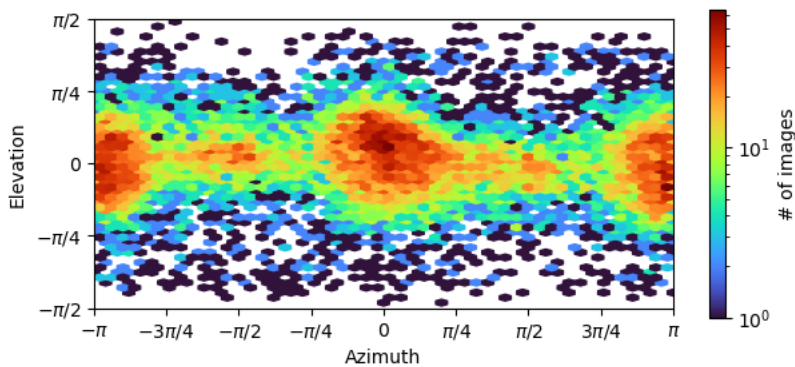
> [2026-06-09 15:29:23.02 GMT-4] Guinier Plot Iteration 003



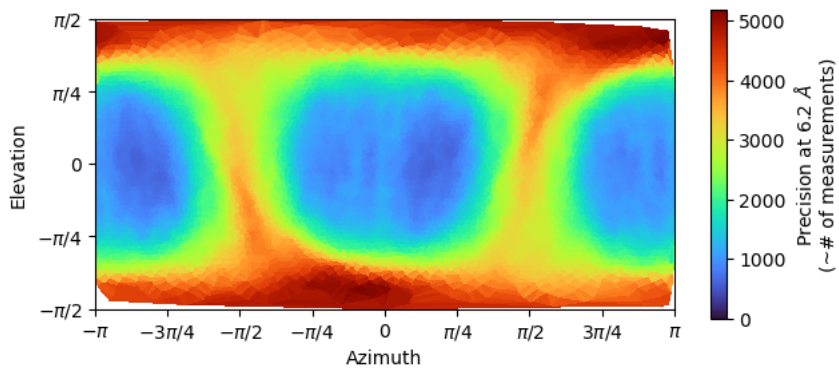
> [2026-06-09 15:29:23.38 GMT-4] Noise Model Iteration 003



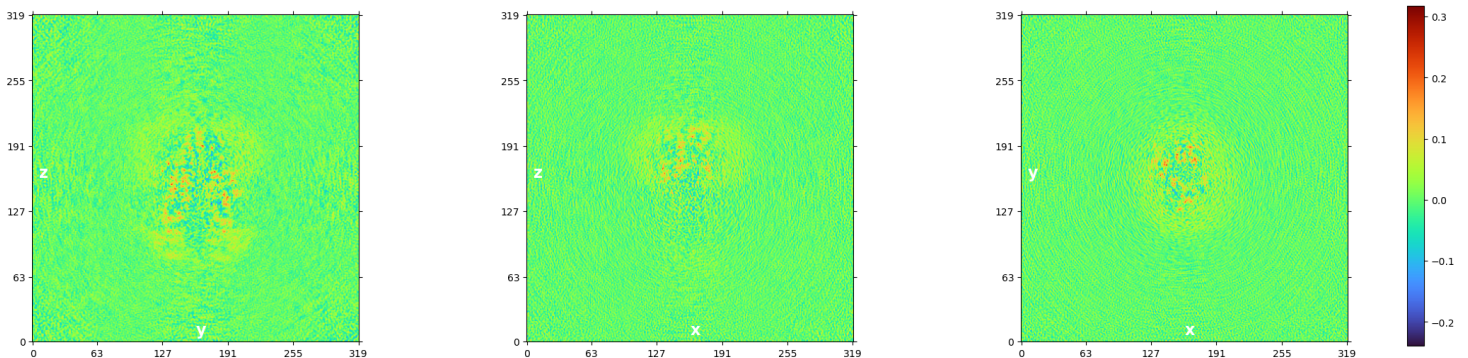
> [2026-06-09 15:29:24.26 GMT-4] Viewing Direction Distribution Iteration 003



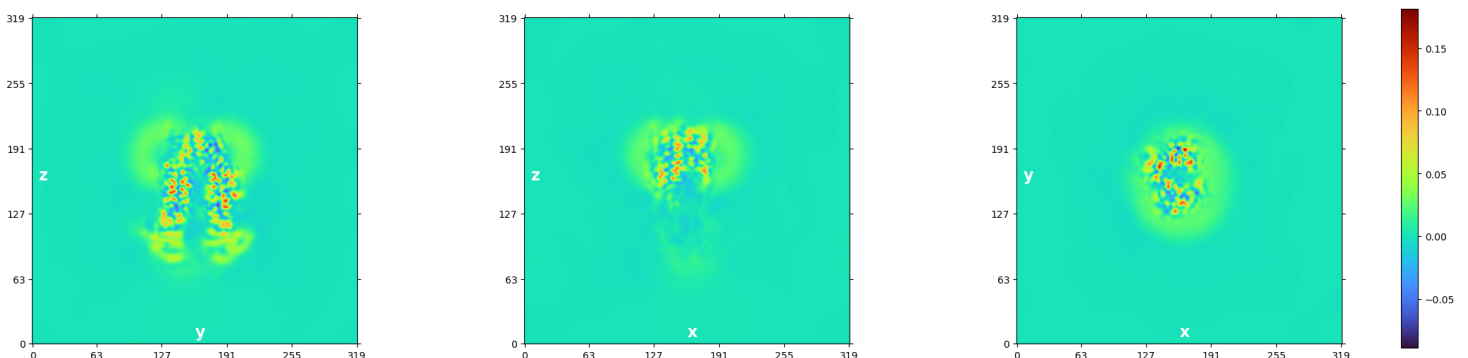
> [2026-06-09 15:29:25.68 GMT-4] Posterior Precision Directional Distribution Iteration 003



```
> [2026-06-09 15:29:25.68 GMT-4] [CPU: 11.31 GB] [Avail: 301.81 GB] Done in 20.072s.
> [2026-06-09 15:29:25.68 GMT-4] [CPU: 11.31 GB] [Avail: 301.80 GB] Outputting files..
> [2026-06-09 15:29:41.73 GMT-4] [CPU: 11.19 GB] [Avail: 299.37 GB] Done in 16.041s.
> [2026-06-09 15:29:41.73 GMT-4] [CPU: 11.19 GB] [Avail: 299.37 GB] Done iteration 3 in 180.242s. Total time so far 730.940s
> [2026-06-09 15:29:42.55 GMT-4] [CPU: 11.19 GB] [Avail: 300.88 GB] ----- Start Iteration 4
> [2026-06-09 15:29:42.55 GMT-4] [CPU: 11.19 GB] [Avail: 300.87 GB] Using Max Alignment Radius 68.698 (3.866Å)
> [2026-06-09 15:29:42.56 GMT-4] [CPU: 11.19 GB] [Avail: 300.87 GB] Auto batchsize: 7972 in each split
> [2026-06-09 15:29:42.56 GMT-4] [CPU: 11.19 GB] [Avail: 300.86 GB] Using dynamic mask.
> [2026-06-09 15:30:14.14 GMT-4] [CPU: 11.19 GB] [Avail: 298.48 GB] -- THR 0 BATCH 500 NUM 1986 TOTAL 5.7233369 ELAPSED 37.296465
--
> [2026-06-09 15:30:54.09 GMT-4] [CPU: 11.95 GB] [Avail: 298.32 GB] Processed 15925.000 images in 39.945s.
> [2026-06-09 15:30:57.54 GMT-4] [CPU: 11.98 GB] [Avail: 298.31 GB] Computing FSCs...
> [2026-06-09 15:30:57.55 GMT-4] [CPU: 11.98 GB] [Avail: 298.31 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:30:57.55 GMT-4] [CPU: 11.98 GB] [Avail: 298.31 GB] Computing FFTs on GPU.
> [2026-06-09 15:30:59.47 GMT-4] [CPU: 11.98 GB] [Avail: 299.60 GB] Done in 1.926s
> [2026-06-09 15:30:59.48 GMT-4] [CPU: 11.98 GB] [Avail: 299.59 GB] Computing cFSCs...
> [2026-06-09 15:31:01.58 GMT-4] [CPU: 11.98 GB] [Avail: 299.47 GB] Done in 2.099s
> [2026-06-09 15:31:01.59 GMT-4] [CPU: 11.98 GB] [Avail: 299.47 GB] Using Filter Radius 74.022 (3.588Å) | Previous: 68.698
(3.866Å)
> [2026-06-09 15:31:08.54 GMT-4] [CPU: 11.22 GB] [Avail: 300.81 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:31:08.54 GMT-4] [CPU: 11.22 GB] [Avail: 300.80 GB] Running local cross validation for A ...
> [2026-06-09 15:31:31.52 GMT-4] [CPU: 11.44 GB] [Avail: 298.00 GB] Local cross validation A done in 22.973s
> [2026-06-09 15:31:32.95 GMT-4] FSC Filtered Side A
```



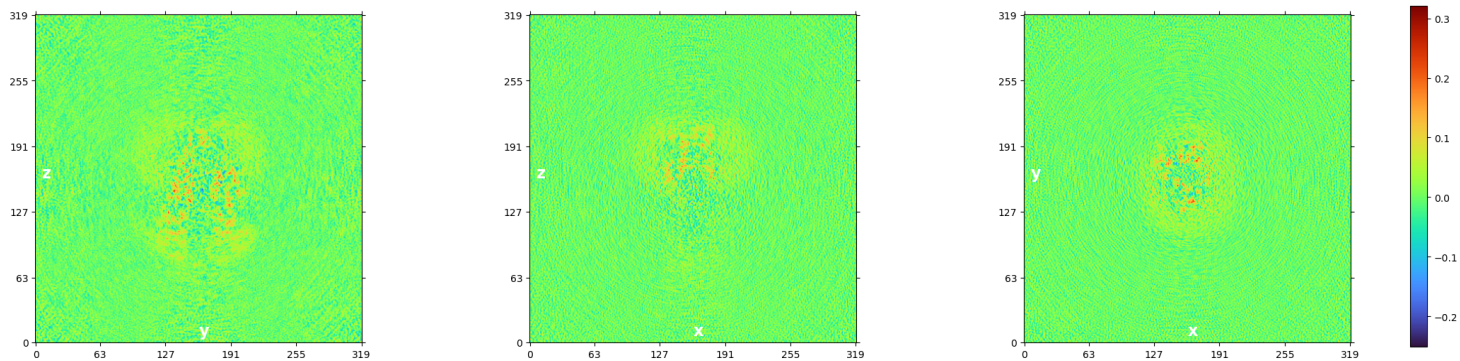
> [2026-06-09 15:31:33.98 GMT-4] CV Filtered Side A



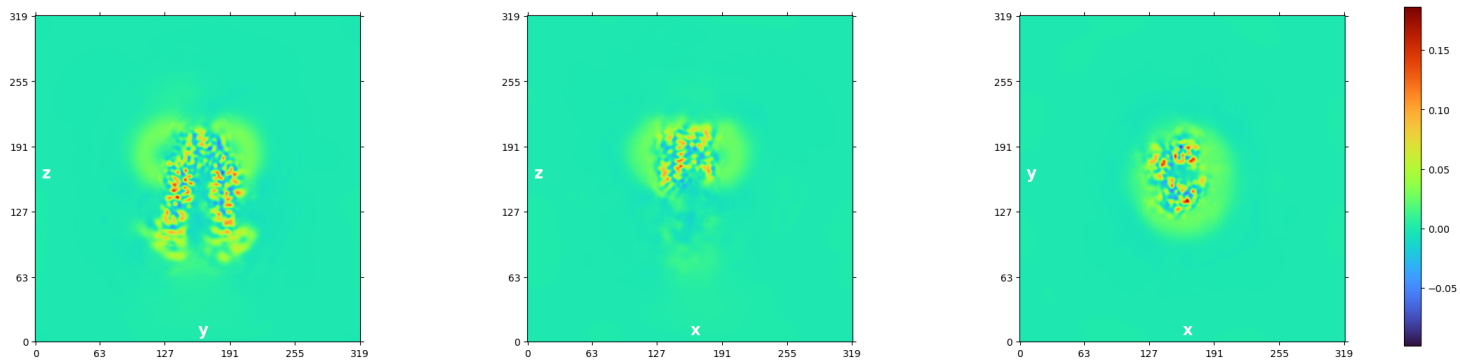
> [2026-06-09 15:31:33.98 GMT-4] [CPU: 11.44 GB] [Avail: 297.11 GB] Running local cross validation for B ...

> [2026-06-09 15:31:56.60 GMT-4] [CPU: 11.69 GB] [Avail: 294.57 GB] Local cross validation B done in 22.613s

> [2026-06-09 15:31:58.00 GMT-4] FSC Filtered Side B



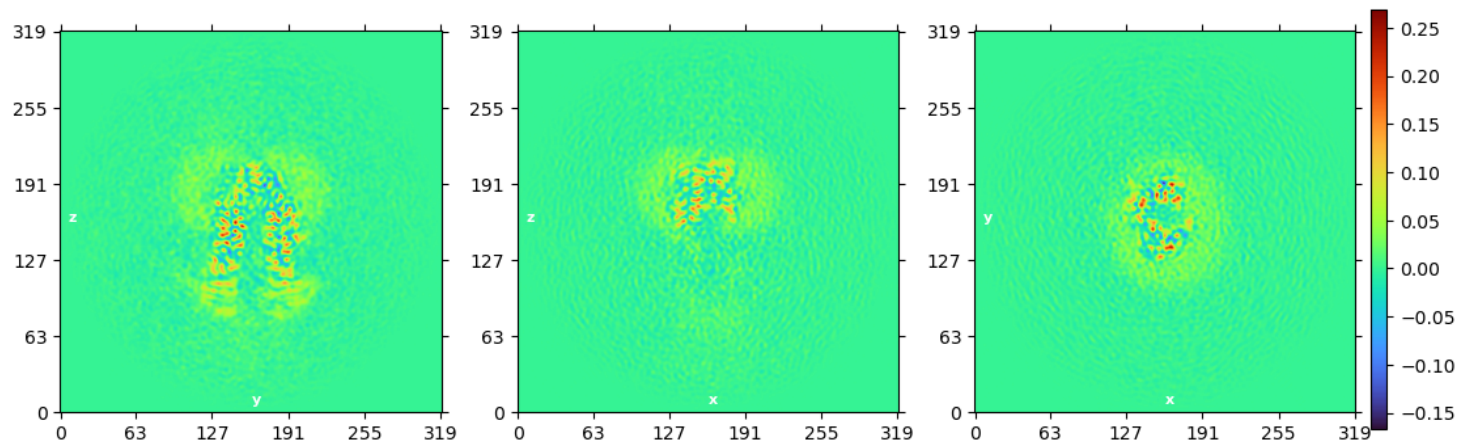
> [2026-06-09 15:31:59.02 GMT-4] CV Filtered Side B



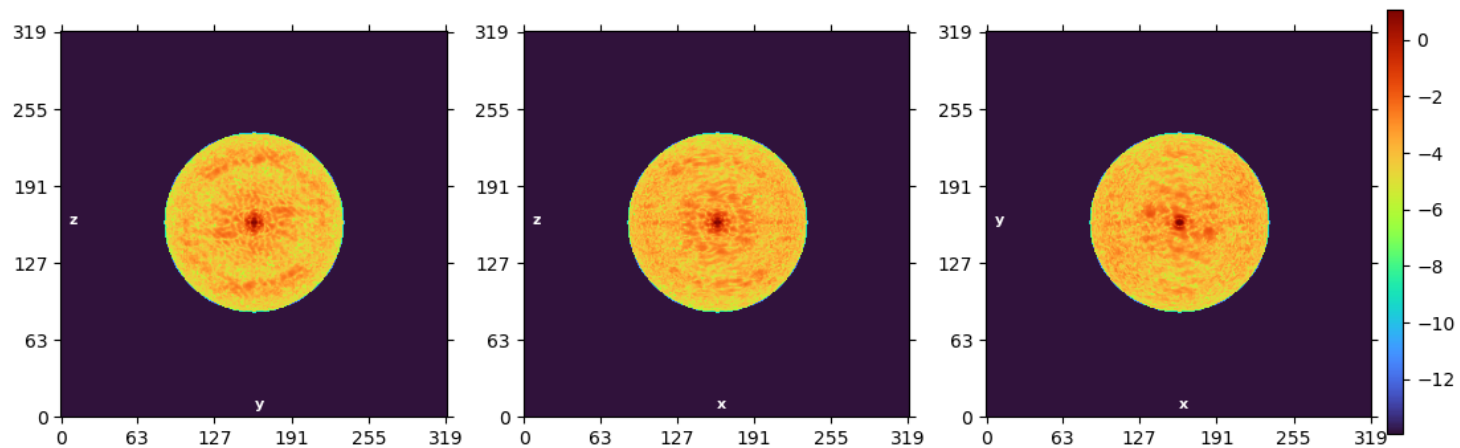
> [2026-06-09 15:32:09.30 GMT-4] [CPU: 11.73 GB] [Avail: 292.34 GB] Estimated Bfactor: -68.3

> [2026-06-09 15:32:09.31 GMT-4] [CPU: 11.73 GB] [Avail: 292.34 GB] Plotting..

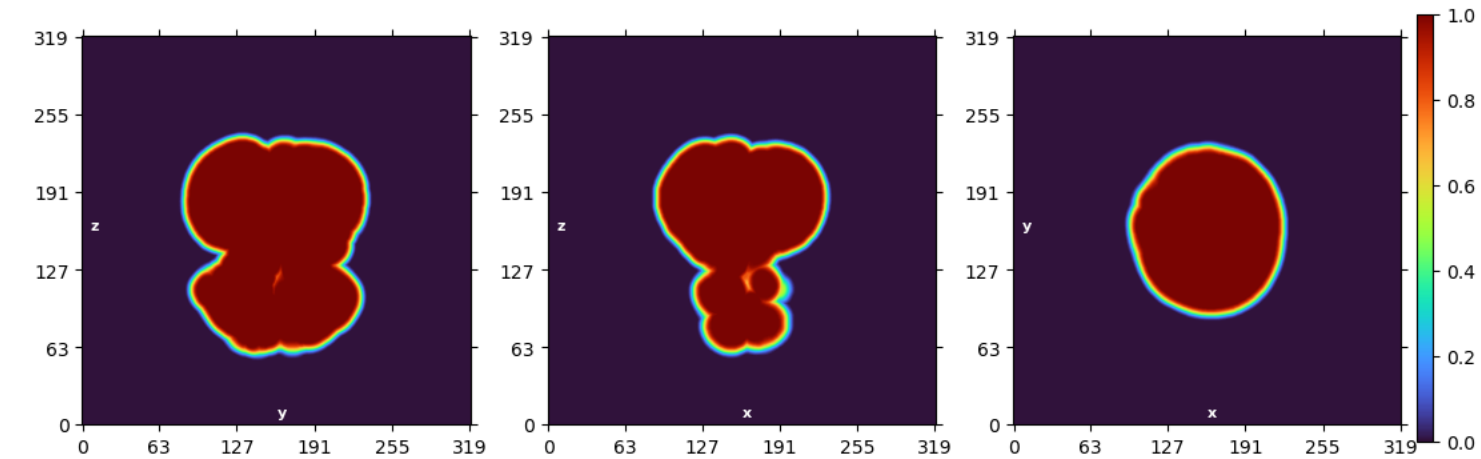
> [2026-06-09 15:32:15.76 GMT-4] Real Space Slices Iteration 004



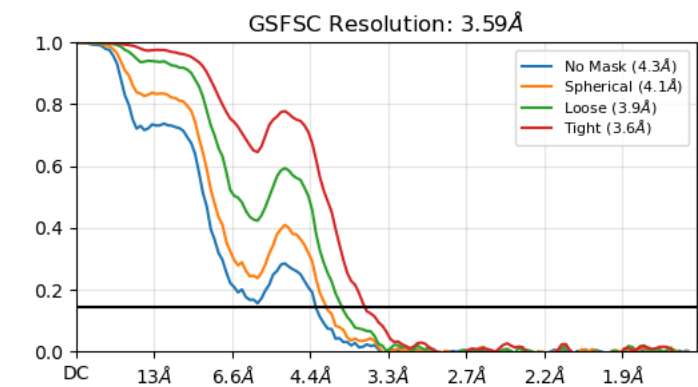
> [2026-06-09 15:32:17.06 GMT-4] Fourier Space Slices Iteration 004



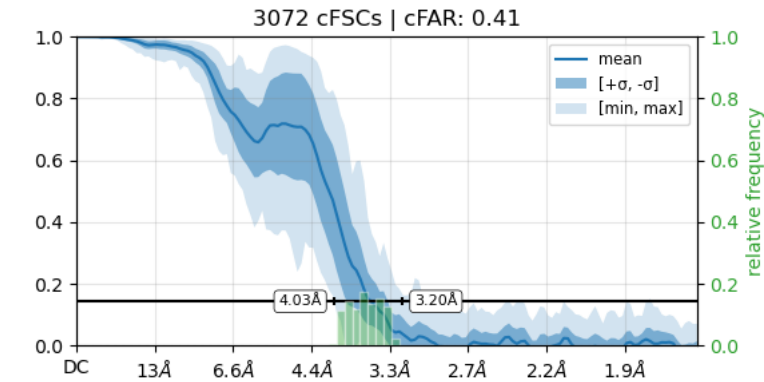
> [2026-06-09 15:32:19.05 GMT-4] Real Space Mask Slices Iteration 004



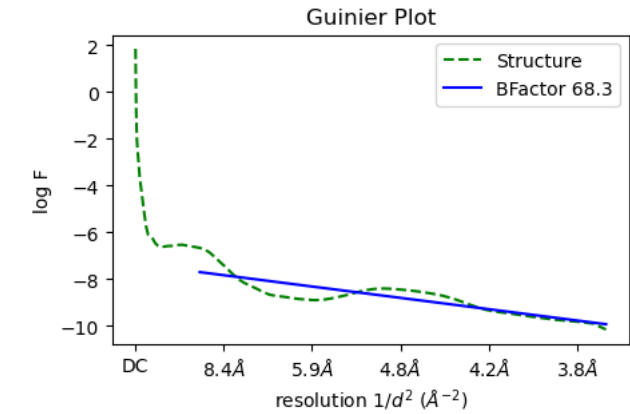
> [2026-06-09 15:32:19.60 GMT-4] FSC Iteration 004



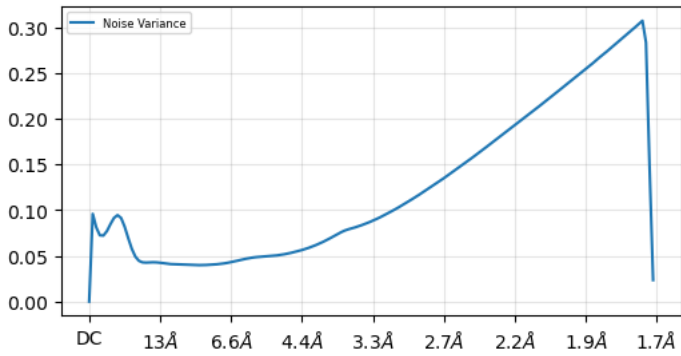
> [2026-06-09 15:32:20.32 GMT-4] cFSCs (Half-angle: 20°) Iteration 004, with tight mask



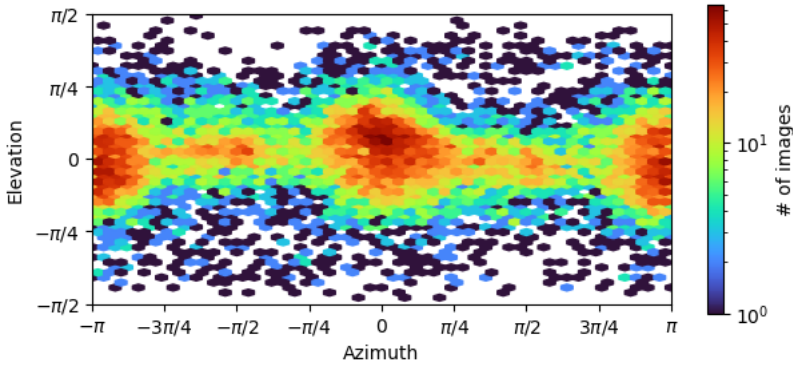
> [2026-06-09 15:32:26.30 GMT-4] Guinier Plot Iteration 004



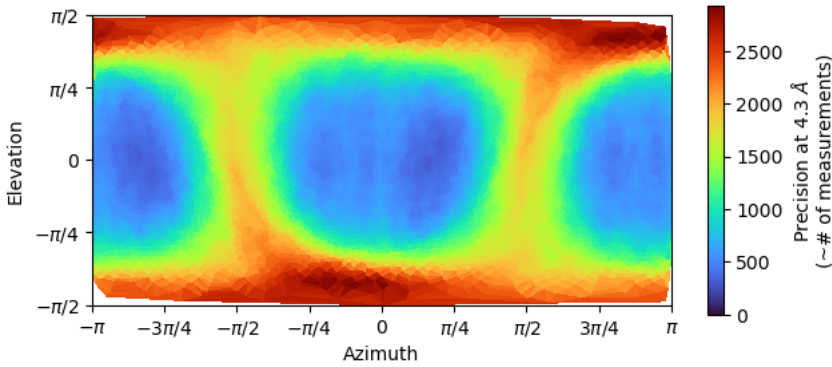
> [2026-06-09 15:32:26.79 GMT-4] Noise Model Iteration 004



> [2026-06-09 15:32:27.83 GMT-4] Viewing Direction Distribution Iteration 004

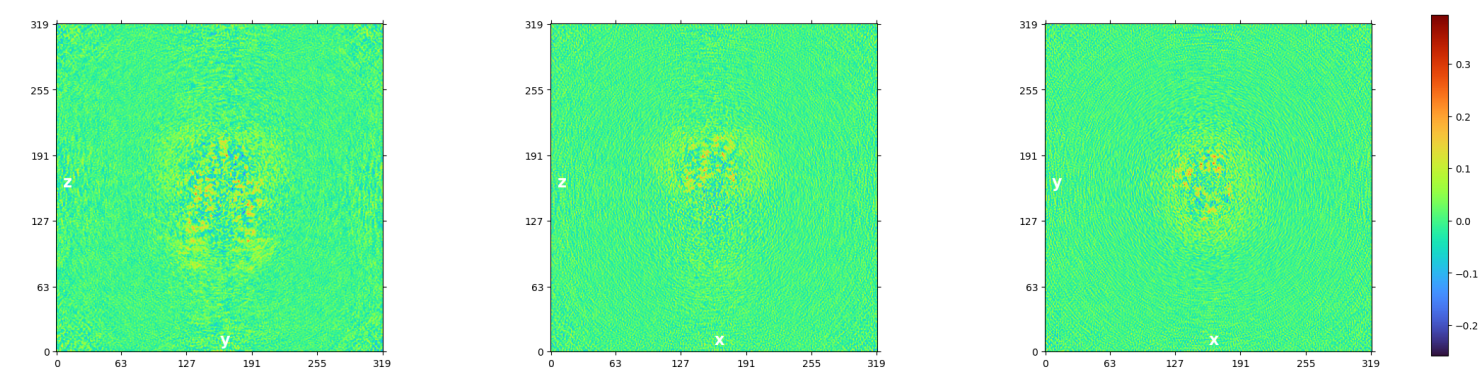


> [2026-06-09 15:32:29.68 GMT-4] Posterior Precision Directional Distribution Iteration 004

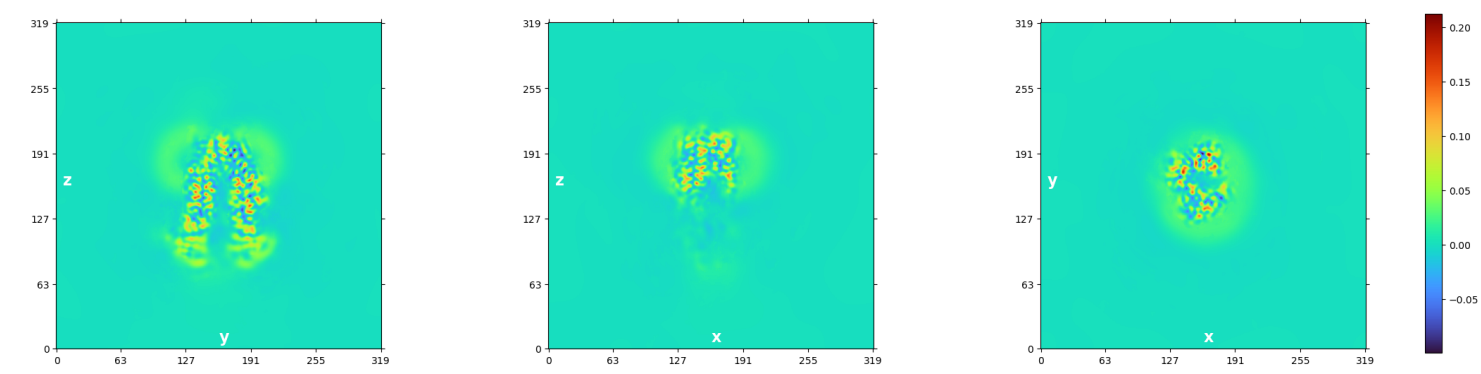


```
> [2026-06-09 15:32:29.68 GMT-4] [CPU: 11.73 GB] [Avail: 289.21 GB] Done in 20.370s.
> [2026-06-09 15:32:29.68 GMT-4] [CPU: 11.73 GB] [Avail: 289.21 GB] Outputting files..
> [2026-06-09 15:32:46.25 GMT-4] [CPU: 11.63 GB] [Avail: 290.25 GB] Done in 16.557s.
> [2026-06-09 15:32:46.25 GMT-4] [CPU: 11.63 GB] [Avail: 290.28 GB] Done iteration 4 in 183.694s. Total time so far 915.459s
> [2026-06-09 15:32:47.07 GMT-4] [CPU: 11.63 GB] [Avail: 289.78 GB] ----- Start Iteration 5
> [2026-06-09 15:32:47.07 GMT-4] [CPU: 11.63 GB] [Avail: 289.77 GB] Using Max Alignment Radius 74.022 (3.588A)
> [2026-06-09 15:32:47.08 GMT-4] [CPU: 11.63 GB] [Avail: 289.76 GB] Auto batchsize: 7972 in each split
> [2026-06-09 15:32:47.10 GMT-4] [CPU: 11.63 GB] [Avail: 289.71 GB] Using dynamic mask.
> [2026-06-09 15:33:20.91 GMT-4] [CPU: 11.63 GB] [Avail: 287.90 GB] -- THR 0 BATCH 500 NUM 1986 TOTAL 5.9244346 ELAPSED 37.546843
--
> [2026-06-09 15:34:00.71 GMT-4] [CPU: 12.39 GB] [Avail: 287.81 GB] Processed 15925.000 images in 39.792s.
> [2026-06-09 15:34:03.83 GMT-4] [CPU: 12.42 GB] [Avail: 287.28 GB] Computing FSCs...
> [2026-06-09 15:34:03.84 GMT-4] [CPU: 12.42 GB] [Avail: 287.27 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:34:03.84 GMT-4] [CPU: 12.42 GB] [Avail: 287.26 GB] Computing FFTs on GPU.
> [2026-06-09 15:34:05.69 GMT-4] [CPU: 12.42 GB] [Avail: 286.82 GB] Done in 1.857s
> [2026-06-09 15:34:05.69 GMT-4] [CPU: 12.42 GB] [Avail: 286.81 GB] Computing cFSCs...
> [2026-06-09 15:34:07.50 GMT-4] [CPU: 12.42 GB] [Avail: 287.70 GB] Done in 1.804s
> [2026-06-09 15:34:07.51 GMT-4] [CPU: 12.42 GB] [Avail: 287.69 GB] Using Filter Radius 76.734 (3.461A) | Previous: 74.022
(3.588A)
> [2026-06-09 15:34:13.60 GMT-4] [CPU: 11.66 GB] [Avail: 287.73 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:34:13.61 GMT-4] [CPU: 11.66 GB] [Avail: 287.72 GB] Running local cross validation for A ...
> [2026-06-09 15:34:36.91 GMT-4] [CPU: 11.88 GB] [Avail: 288.44 GB] Local cross validation A done in 23.301s
```

> [2026-06-09 15:34:38.15 GMT-4] FSC Filtered Side A



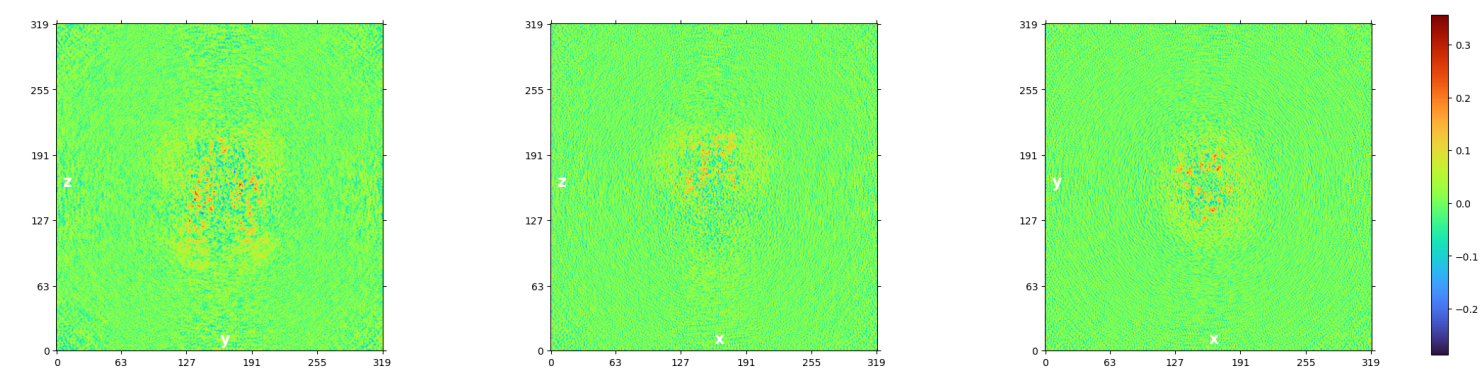
> [2026-06-09 15:34:39.15 GMT-4] CV Filtered Side A



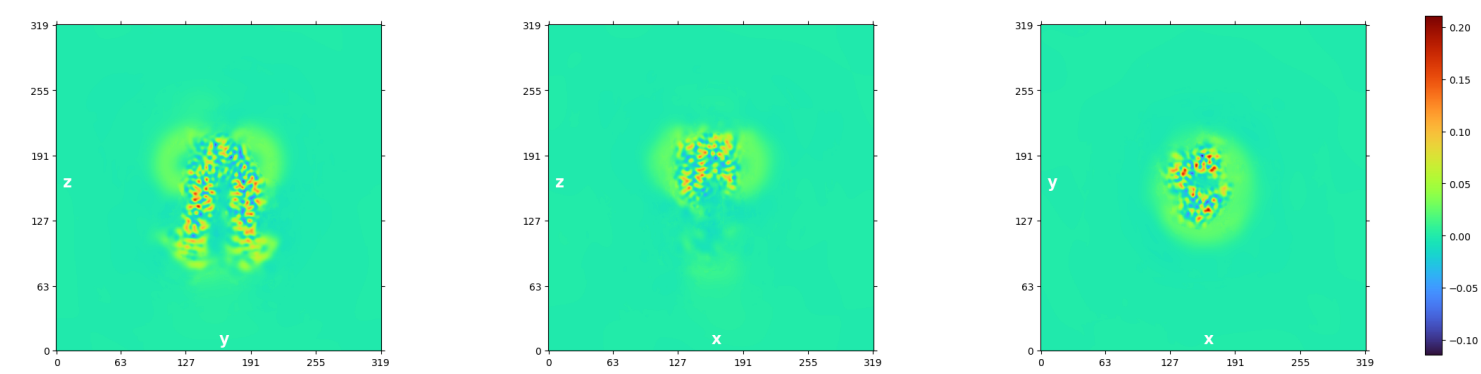
> [2026-06-09 15:34:39.15 GMT-4] [CPU: 11.88 GB] [Avail: 288.02 GB] Running local cross validation for B ...

> [2026-06-09 15:35:02.82 GMT-4] [CPU: 12.13 GB] [Avail: 298.16 GB] Local cross validation B done in 23.661s

> [2026-06-09 15:35:04.23 GMT-4] FSC Filtered Side B



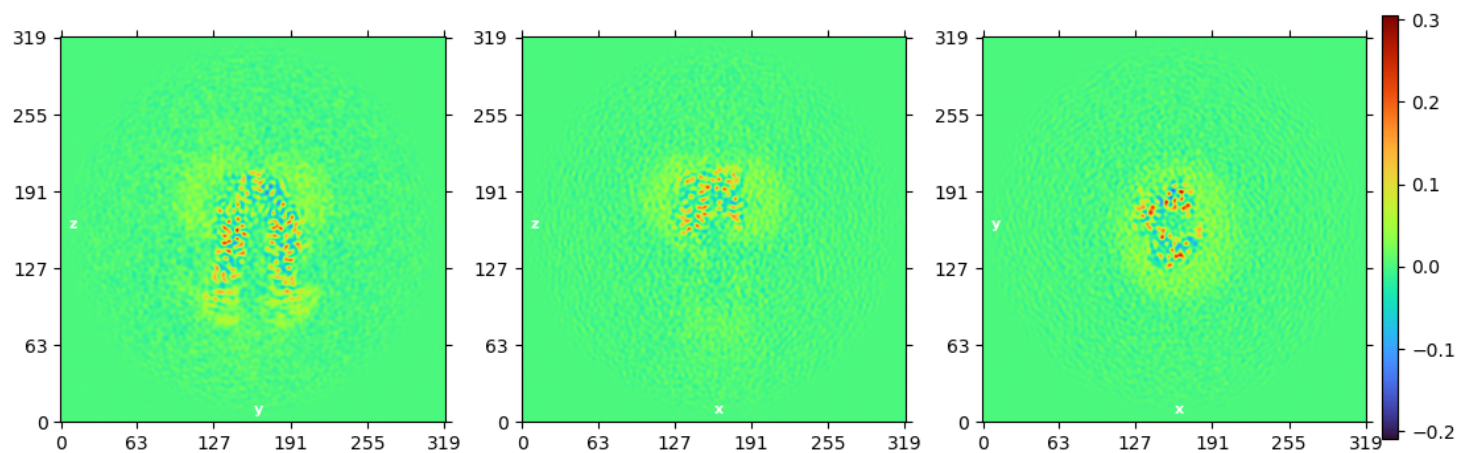
> [2026-06-09 15:35:05.17 GMT-4] CV Filtered Side B



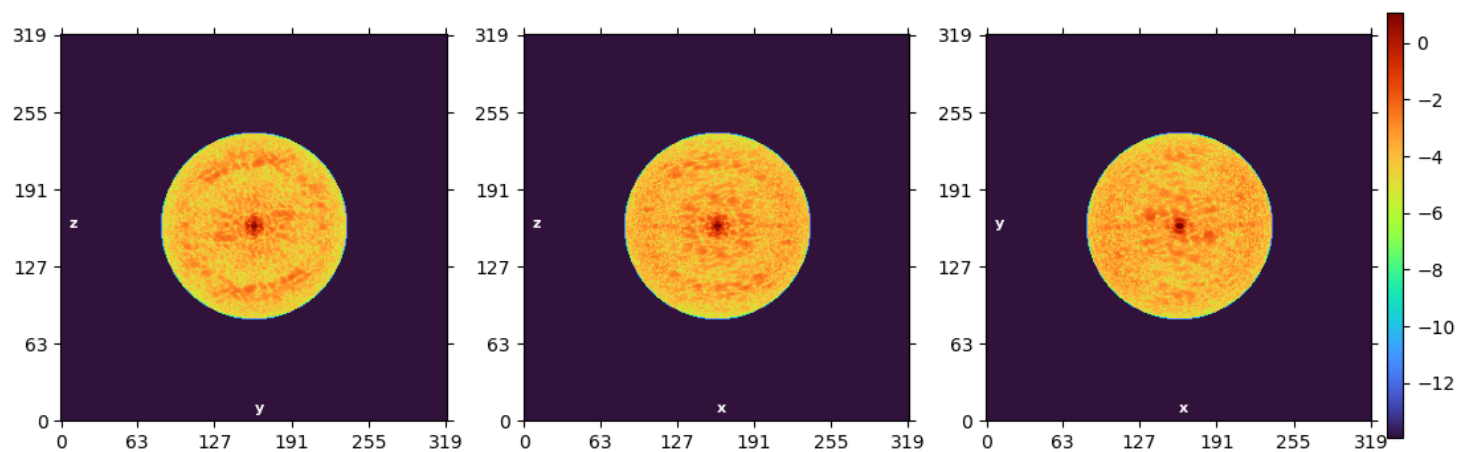
> [2026-06-09 15:35:16.48 GMT-4] [CPU: 12.19 GB] [Avail: 296.31 GB] Estimated Bfactor: -62.4

> [2026-06-09 15:35:16.49 GMT-4] [CPU: 12.19 GB] [Avail: 296.31 GB] Plotting..

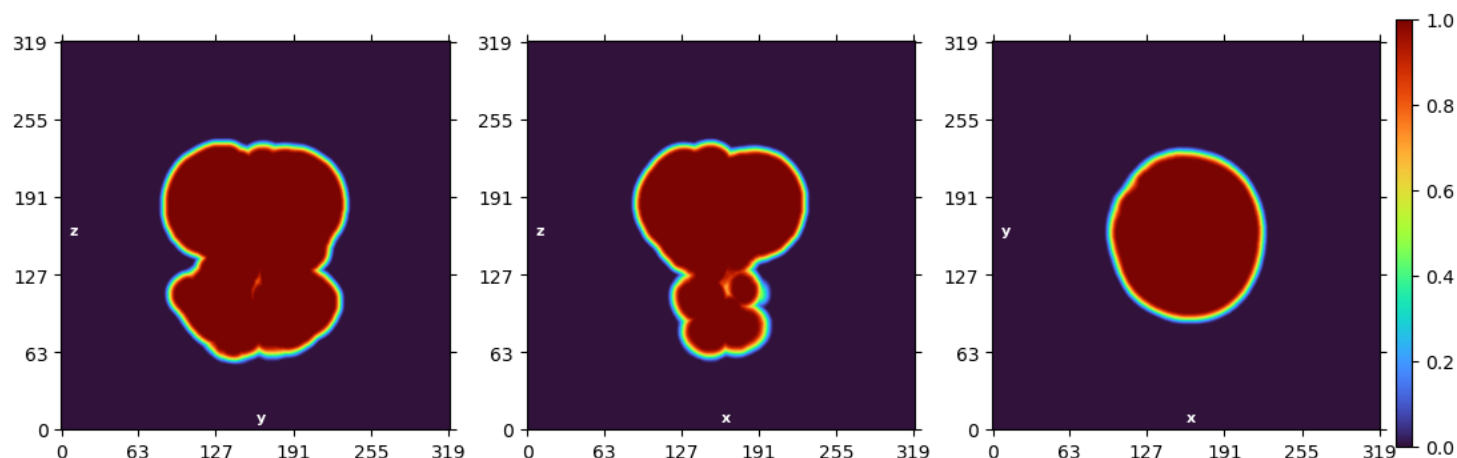
> [2026-06-09 15:35:23.63 GMT-4] Real Space Slices Iteration 005



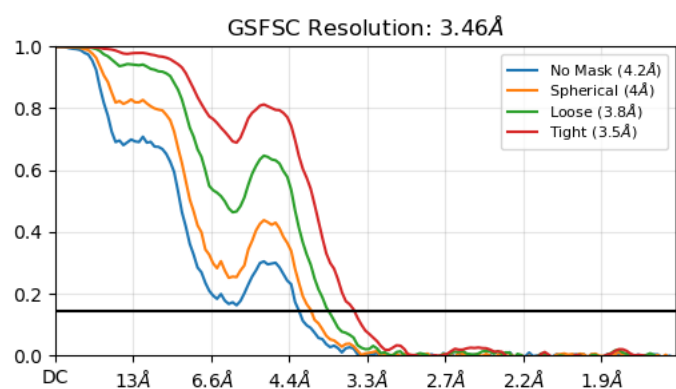
> [2026-06-09 15:35:25.08 GMT-4] Fourier Space Slices Iteration 005



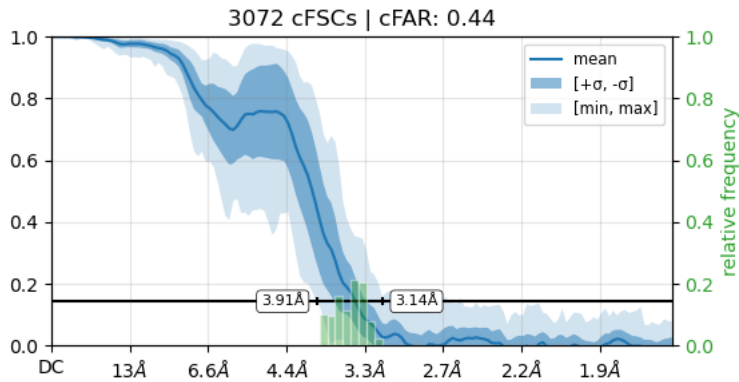
> [2026-06-09 15:35:27.22 GMT-4] Real Space Mask Slices Iteration 005



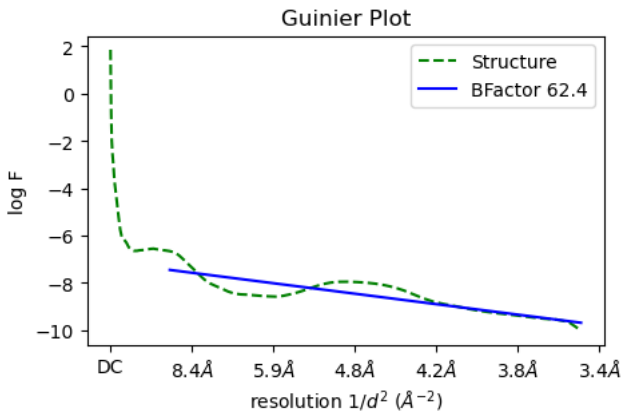
> [2026-06-09 15:35:27.69 GMT-4] FSC Iteration 005



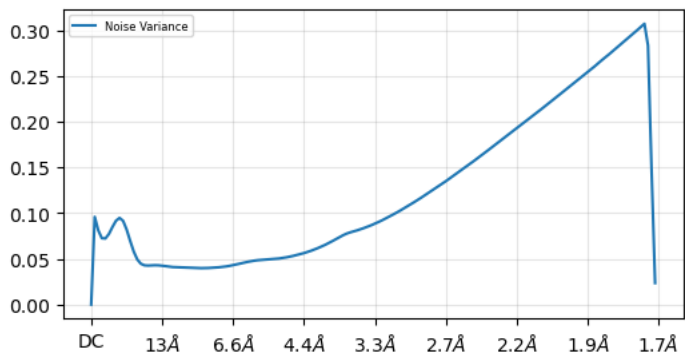
> [2026-06-09 15:35:28.50 GMT-4] cFSCs (Half-angle: 20°) Iteration 005, with tight mask



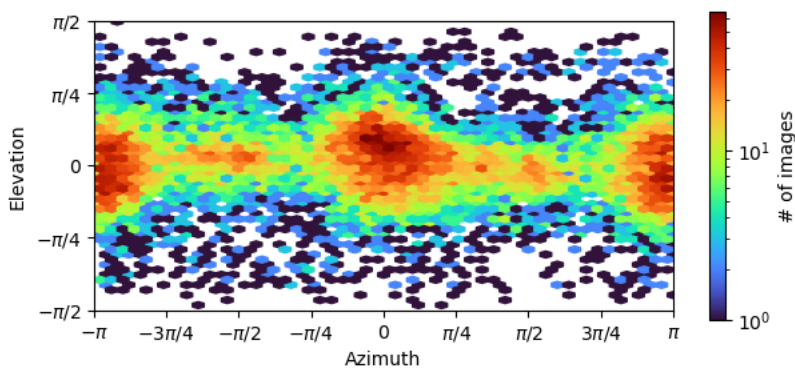
> [2026-06-09 15:35:34.84 GMT-4] Guinier Plot Iteration 005



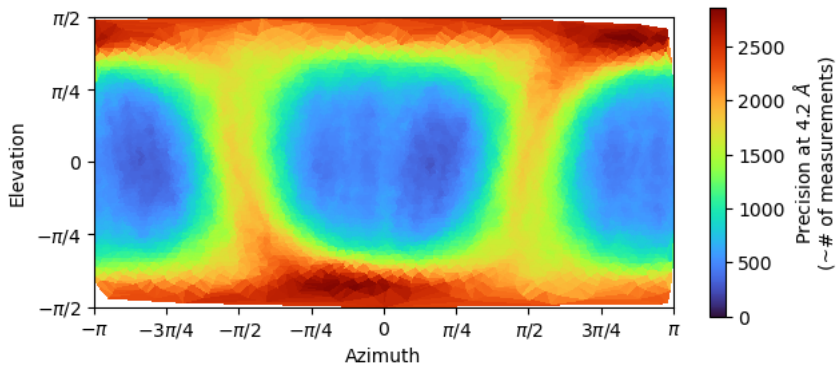
> [2026-06-09 15:35:35.21 GMT-4] Noise Model Iteration 005



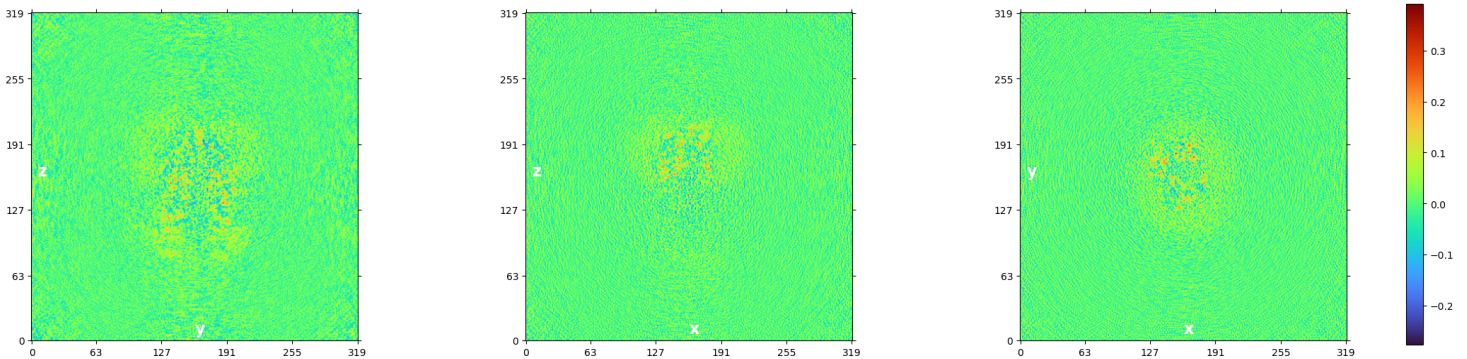
> [2026-06-09 15:35:36.18 GMT-4] Viewing Direction Distribution Iteration 005



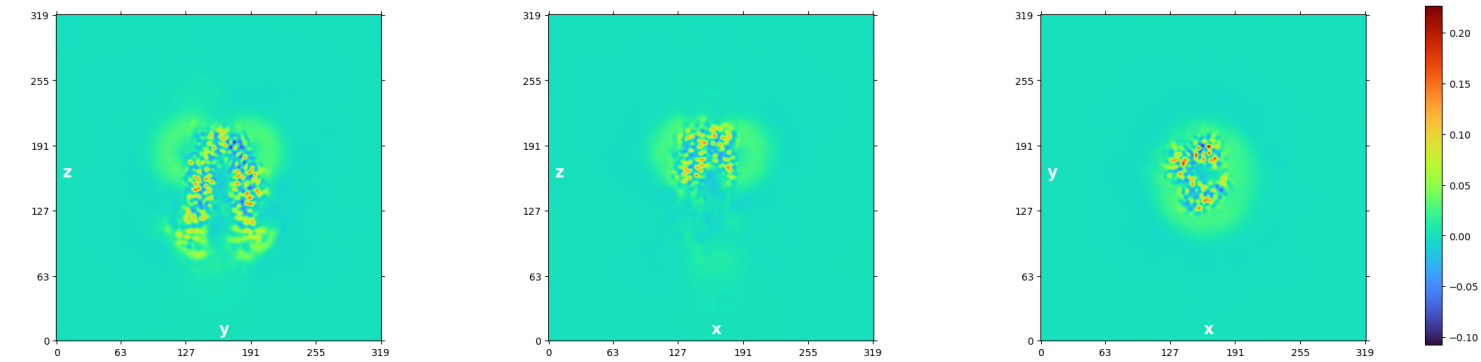
> [2026-06-09 15:35:38.05 GMT-4] Posterior Precision Directional Distribution Iteration 005



```
> [2026-06-09 15:35:38.05 GMT-4] [CPU: 12.19 GB] [Avail: 299.65 GB] Done in 21.563s.
> [2026-06-09 15:35:38.06 GMT-4] [CPU: 12.19 GB] [Avail: 299.65 GB] Outputting files..
> [2026-06-09 15:35:54.45 GMT-4] [CPU: 12.07 GB] [Avail: 295.83 GB] Done in 16.389s.
> [2026-06-09 15:35:54.45 GMT-4] [CPU: 12.07 GB] [Avail: 295.83 GB] Done iteration 5 in 187.379s. Total time so far 1103.661s
> [2026-06-09 15:35:55.35 GMT-4] [CPU: 12.07 GB] [Avail: 296.54 GB] ----- Start Iteration 6
> [2026-06-09 15:35:55.35 GMT-4] [CPU: 12.07 GB] [Avail: 296.54 GB] Using Max Alignment Radius 76.734 (3.461A)
> [2026-06-09 15:35:55.35 GMT-4] [CPU: 12.07 GB] [Avail: 296.53 GB] Using Full Dataset (split 7953 in A, 7972 in B)
> [2026-06-09 15:35:55.36 GMT-4] [CPU: 12.07 GB] [Avail: 296.53 GB] Using dynamic mask.
> [2026-06-09 15:36:28.41 GMT-4] [CPU: 12.07 GB] [Avail: 291.30 GB] -- THR 1 BATCH 500 NUM 1986 TOTAL 4.8687775 ELAPSED 37.944378
--
> [2026-06-09 15:37:08.69 GMT-4] [CPU: 12.82 GB] [Avail: 289.76 GB] Processed 15925.000 images in 40.280s.
> [2026-06-09 15:37:12.21 GMT-4] [CPU: 12.86 GB] [Avail: 291.02 GB] Computing FSCs...
> [2026-06-09 15:37:12.21 GMT-4] [CPU: 12.86 GB] [Avail: 291.01 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:37:12.22 GMT-4] [CPU: 12.86 GB] [Avail: 291.00 GB] Computing FFTs on GPU.
> [2026-06-09 15:37:15.74 GMT-4] [CPU: 12.86 GB] [Avail: 290.81 GB] Done in 3.526s
> [2026-06-09 15:37:15.74 GMT-4] [CPU: 12.86 GB] [Avail: 290.81 GB] Optimizing FSC Mask...
> [2026-06-09 15:37:15.75 GMT-4] [CPU: 12.86 GB] [Avail: 290.82 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:37:15.75 GMT-4] [CPU: 12.86 GB] [Avail: 290.81 GB] Computing FFTs on GPU.
> [2026-06-09 15:37:17.79 GMT-4] [CPU: 13.11 GB] [Avail: 291.27 GB] Done in 2.045s
> [2026-06-09 15:37:17.80 GMT-4] [CPU: 13.11 GB] [Avail: 291.27 GB] Computing cFSCs...
> [2026-06-09 15:37:19.95 GMT-4] [CPU: 13.11 GB] [Avail: 291.05 GB] Done in 2.147s
> [2026-06-09 15:37:19.95 GMT-4] [CPU: 13.11 GB] [Avail: 291.05 GB] Using Filter Radius 79.871 (3.325A) | Previous: 76.734
(3.461A)
> [2026-06-09 15:37:26.72 GMT-4] [CPU: 12.35 GB] [Avail: 290.58 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:37:26.72 GMT-4] [CPU: 12.35 GB] [Avail: 290.57 GB] Running local cross validation for A ...
> [2026-06-09 15:37:50.16 GMT-4] [CPU: 12.57 GB] [Avail: 288.96 GB] Local cross validation A done in 23.435s
> [2026-06-09 15:37:51.76 GMT-4] FSC Filtered Side A
```



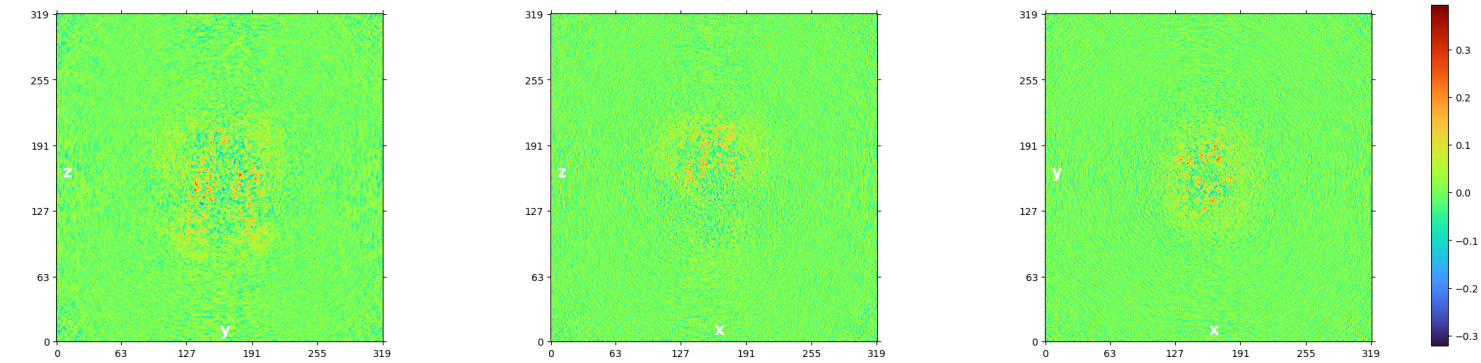
> [2026-06-09 15:37:52.83 GMT-4] CV Filtered Side A



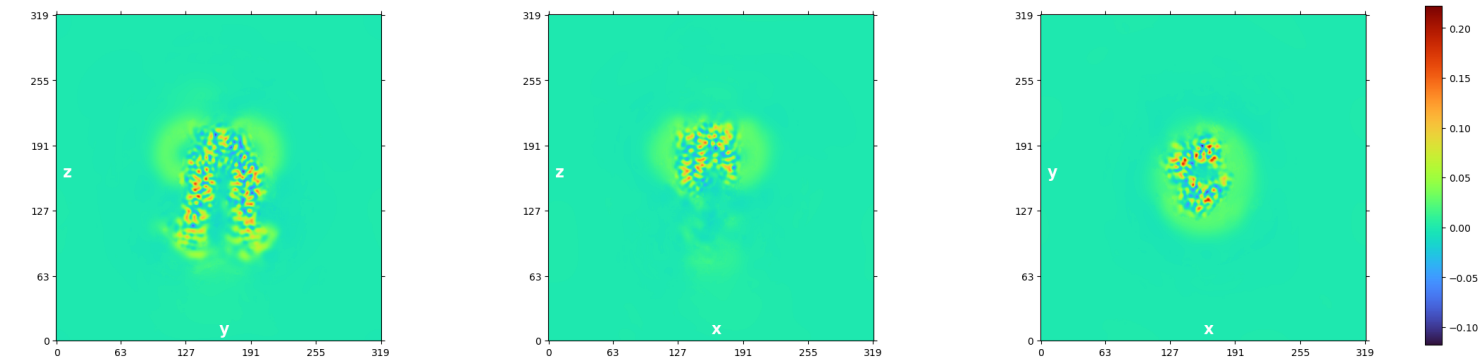
> [2026-06-09 15:37:52.83 GMT-4] [CPU: 12.57 GB] [Avail: 289.73 GB] Running local cross validation for B ...

> [2026-06-09 15:38:17.38 GMT-4] [CPU: 12.82 GB] [Avail: 288.63 GB] Local cross validation B done in 24.542s

> [2026-06-09 15:38:18.83 GMT-4] FSC Filtered Side B



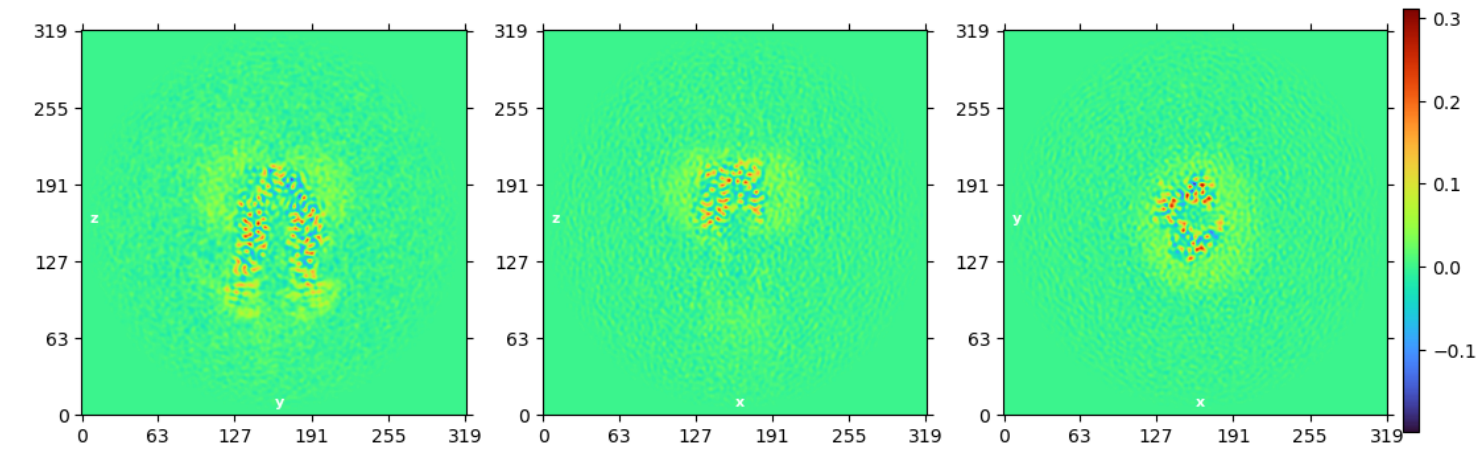
> [2026-06-09 15:38:19.87 GMT-4] CV Filtered Side B



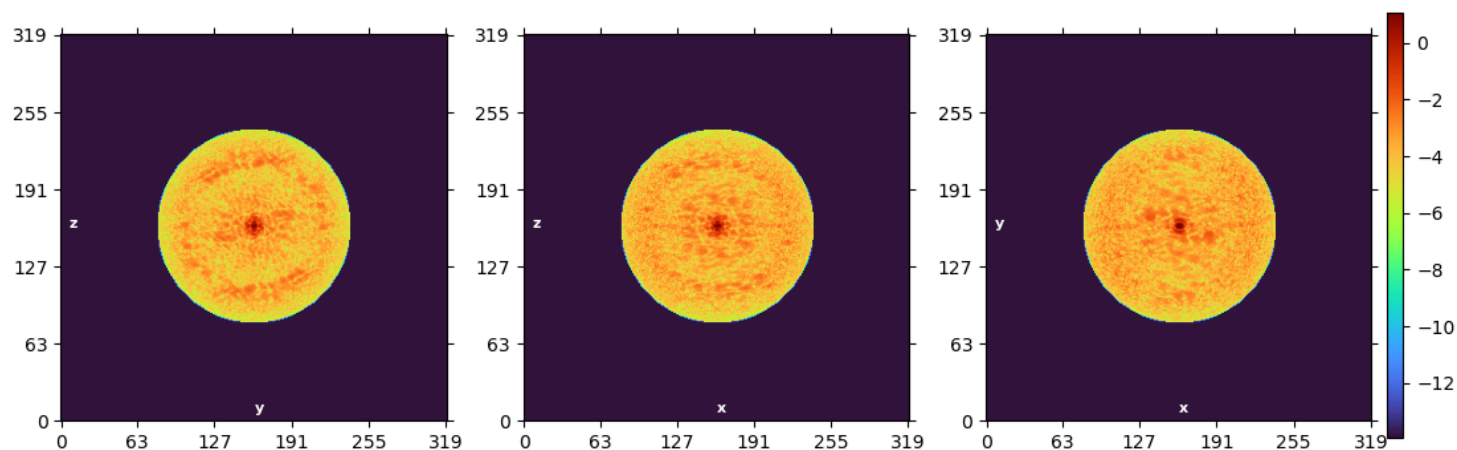
> [2026-06-09 15:38:31.46 GMT-4] [CPU: 12.82 GB] [Avail: 286.86 GB] Estimated Bfactor: -57.6

> [2026-06-09 15:38:31.47 GMT-4] [CPU: 12.82 GB] [Avail: 286.85 GB] Plotting..

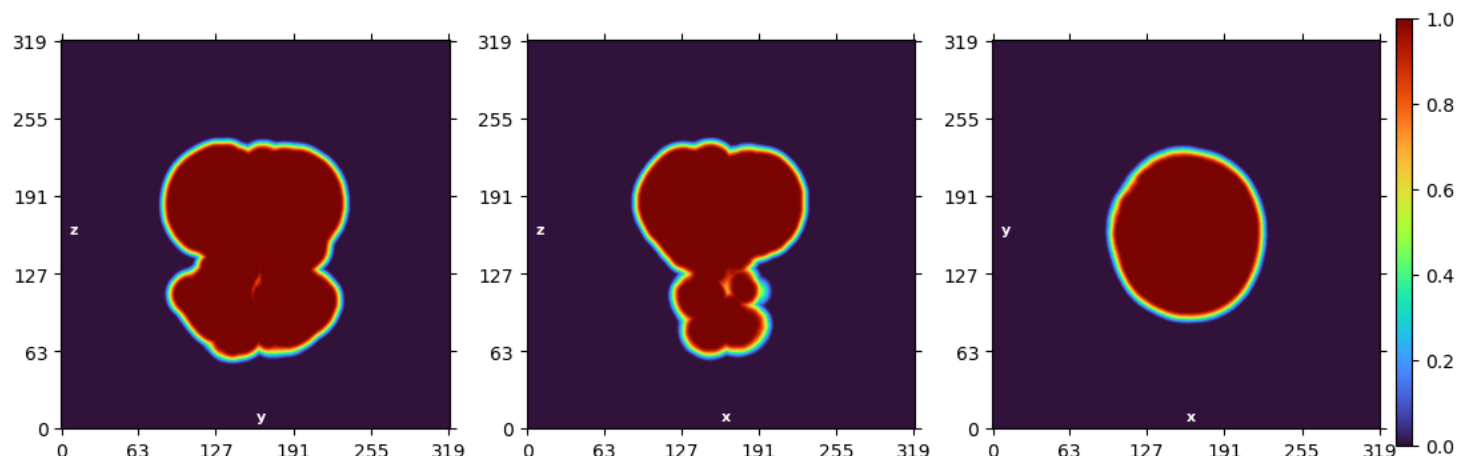
> [2026-06-09 15:38:38.36 GMT-4] Real Space Slices Iteration 006



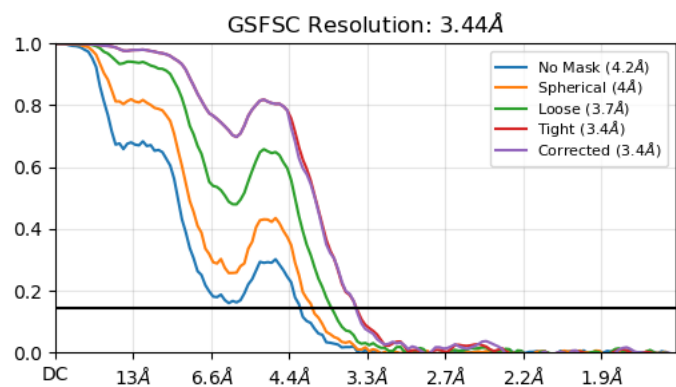
> [2026-06-09 15:38:39.58 GMT-4] Fourier Space Slices Iteration 006



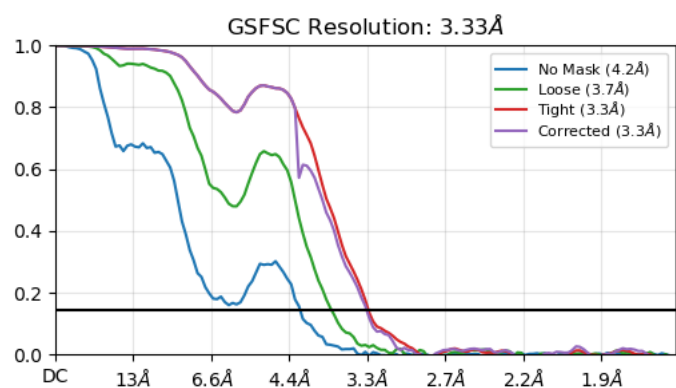
> [2026-06-09 15:38:41.50 GMT-4] Real Space Mask Slices Iteration 006



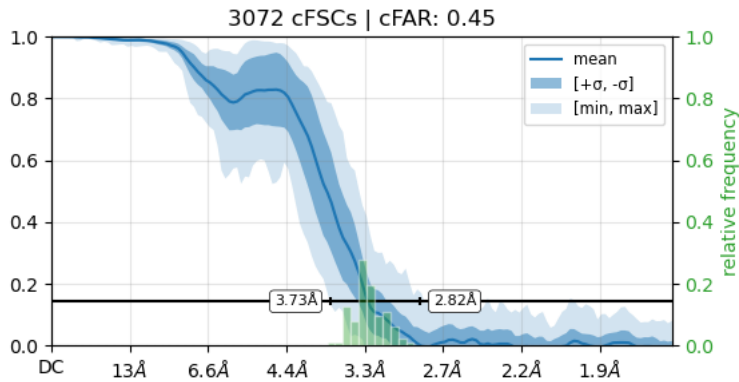
> [2026-06-09 15:38:42.01 GMT-4] FSC Iteration 006



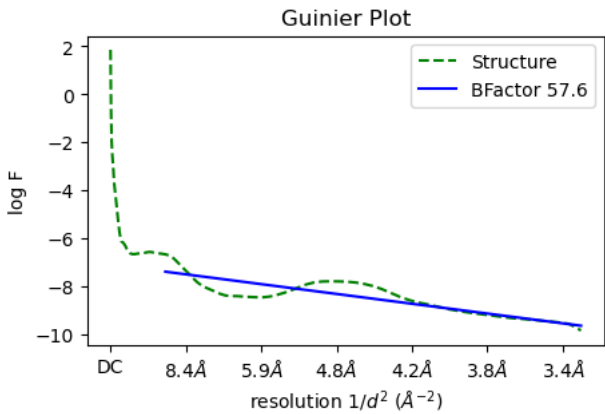
> [2026-06-09 15:38:42.47 GMT-4] FSC Iteration 006, after FSC-mask auto-tightening



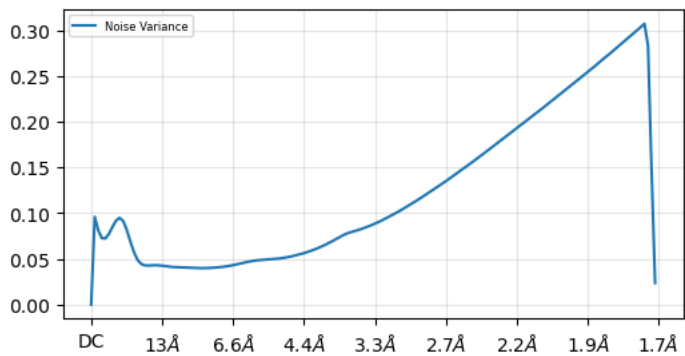
> [2026-06-09 15:38:43.20 GMT-4] cFSCs (Half-angle: 20°) Iteration 006, with autotight mask



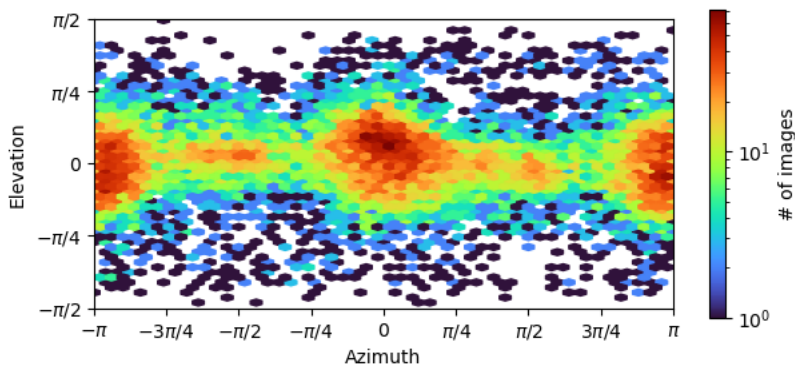
> [2026-06-09 15:38:48.93 GMT-4] Guinier Plot Iteration 006



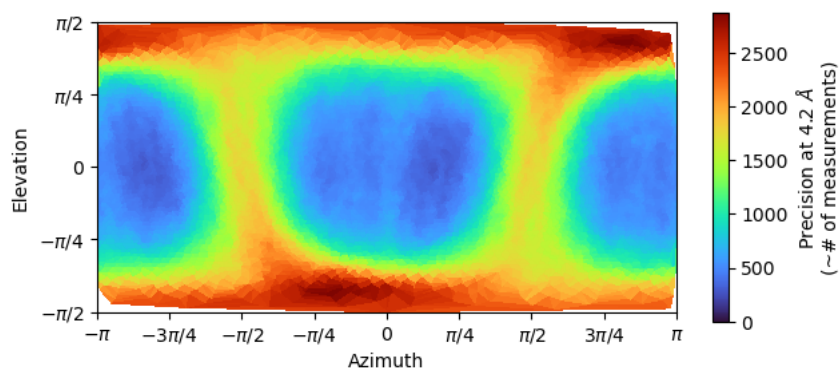
> [2026-06-09 15:38:49.36 GMT-4] Noise Model Iteration 006



> [2026-06-09 15:38:50.20 GMT-4] Viewing Direction Distribution Iteration 006



> [2026-06-09 15:38:51.84 GMT-4] Posterior Precision Directional Distribution Iteration 006



```
> [2026-06-09 15:38:51.85 GMT-4] [CPU: 12.86 GB] [Avail: 285.32 GB] Done in 20.373s.
> [2026-06-09 15:38:51.85 GMT-4] [CPU: 12.86 GB] [Avail: 285.34 GB] Outputting files..
> [2026-06-09 15:39:09.13 GMT-4] [CPU: 12.76 GB] [Avail: 288.55 GB] Done in 17.281s.
> [2026-06-09 15:39:09.14 GMT-4] [CPU: 12.76 GB] [Avail: 288.58 GB] Done iteration 6 in 193.787s. Total time so far 1298.348s
> [2026-06-09 15:39:09.79 GMT-4] [CPU: 12.76 GB] [Avail: 288.41 GB] ===== Done Refinement =====
> [2026-06-09 15:39:09.79 GMT-4] [CPU: 12.76 GB] [Avail: 288.43 GB] Note that the output structure from refinement has been
                                                                    lowpass filtered to the gold-standard FSC resolution
                                                                    estimated
                                                                    above. However, the structure probably needs sharpening in
                                                                    order to best visualize high resolution features. This can be
                                                                    done with the Sharpening task (as a new experiment).

> [2026-06-09 15:39:09.79 GMT-4] [CPU: 12.76 GB] [Avail: 288.45 GB] Full run took 1299.006s
> [2026-06-09 15:39:10.39 GMT-4] [CPU: 6.90 GB] [Avail: 294.00 GB] -----
-
> [2026-06-09 15:39:10.40 GMT-4] [CPU: 6.90 GB] [Avail: 294.00 GB] Compiling job outputs...
> [2026-06-09 15:39:10.40 GMT-4] [CPU: 6.90 GB] [Avail: 293.99 GB] Passing through outputs for output group particles from input
                                                                    group particles
> [2026-06-09 15:39:10.45 GMT-4] [CPU: 6.91 GB] [Avail: 293.92 GB] This job outputted results ['alignments3D', 'ctf']
> [2026-06-09 15:39:10.45 GMT-4] [CPU: 6.91 GB] [Avail: 293.91 GB] Loaded output dset with 15925 items
> [2026-06-09 15:39:10.46 GMT-4] [CPU: 6.91 GB] [Avail: 293.91 GB] Passthrough results ['blob', 'alignments3D_multi']
> [2026-06-09 15:39:10.58 GMT-4] [CPU: 6.91 GB] [Avail: 293.68 GB] Loaded passthrough dset with 15925 items
> [2026-06-09 15:39:10.62 GMT-4] [CPU: 6.91 GB] [Avail: 293.51 GB] Intersection of output and passthrough has 15925 items
> [2026-06-09 15:39:10.67 GMT-4] [CPU: 6.91 GB] [Avail: 293.72 GB] Output dataset contains: ['blob', 'alignments3D_multi']
> [2026-06-09 15:39:10.68 GMT-4] [CPU: 6.91 GB] [Avail: 293.69 GB] Outputting passthrough result blob
> [2026-06-09 15:39:10.69 GMT-4] [CPU: 6.91 GB] [Avail: 293.66 GB] Outputting passthrough result alignments3D_multi
> [2026-06-09 15:39:10.69 GMT-4] [CPU: 6.91 GB] [Avail: 293.72 GB] Checking outputs for output group particles
> [2026-06-09 15:39:10.85 GMT-4] [CPU: 6.91 GB] [Avail: 292.80 GB] Updating job size...
> [2026-06-09 15:39:11.66 GMT-4] [CPU: 6.91 GB] [Avail: 293.43 GB] Exporting job and creating csg files...
> [2026-06-09 15:39:12.47 GMT-4] [CPU: 6.91 GB] [Avail: 293.42 GB] *****
                                                                    **
> [2026-06-09 15:39:12.48 GMT-4] [CPU: 6.91 GB] [Avail: 293.42 GB] Job complete. Total time 1301.72s
```